Gold's Grip: How the Fed Broke the Link

Justin Eloriaga* Caroline Fohlin†

July 2026

Abstract

Gold flows were the main channel of monetary transmission before the age of central banking, yet we still know little about how they moved the economy. We estimate a Bayesian structural VAR on monthly U.S. data for 1890 to 1927 and identify gold shocks with light impact sign and zero restrictions and a set of narrative restrictions across various historical episodes, both before and after the founding of the Federal Reserve, and including both gold inflows and gold outflows. We impose only the impact responses and leave the rest of the path free, and the estimated gold shock still lowers the call rate, lifts equities, and raises output and prices with a lag, so the model's dynamics emerge without being assumed. Transmission is stronger before the founding of the Fed, and the gap is concentrated in the money market and real activity rather than in prices. When we let the dynamics differ by estimating each period on its own subsample, the effect of a gold inflow on the call money rate is about nine times larger before the founding of the Fed, with credible bands that do not overlap, while the response afterward is close to zero. The Federal Reserve appears to have broken the automatic link between outside gold movements and the domestic money market, while leaving the slower response of prices little changed.

JEL C32, D83, E52, E58, E66. **Keywords** Bayesian VAR, narrative sign restrictions, gold standard, monetary transmission, panics.

*International Monetary Fund, Shanghai Center. jeloriaga@imf.org. The views expressed in this paper are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management. We thank Elena Pesavento, Kurt Lunsford, Emi Nakamura, and Juan Rubio-Ramírez, and participants of the 2026 Mountain West Economic History Conference and the Joint 13th CEPR Economic History Symposium and 11th Banco de España Economic History Seminar, for helpful comments.

†Emory University, CEPR, SAFE-Frankfurt, and CESifo. cfohlin@emory.edu.

1 Introduction

Gold is money. Everything else is credit. — John Pierpont Morgan

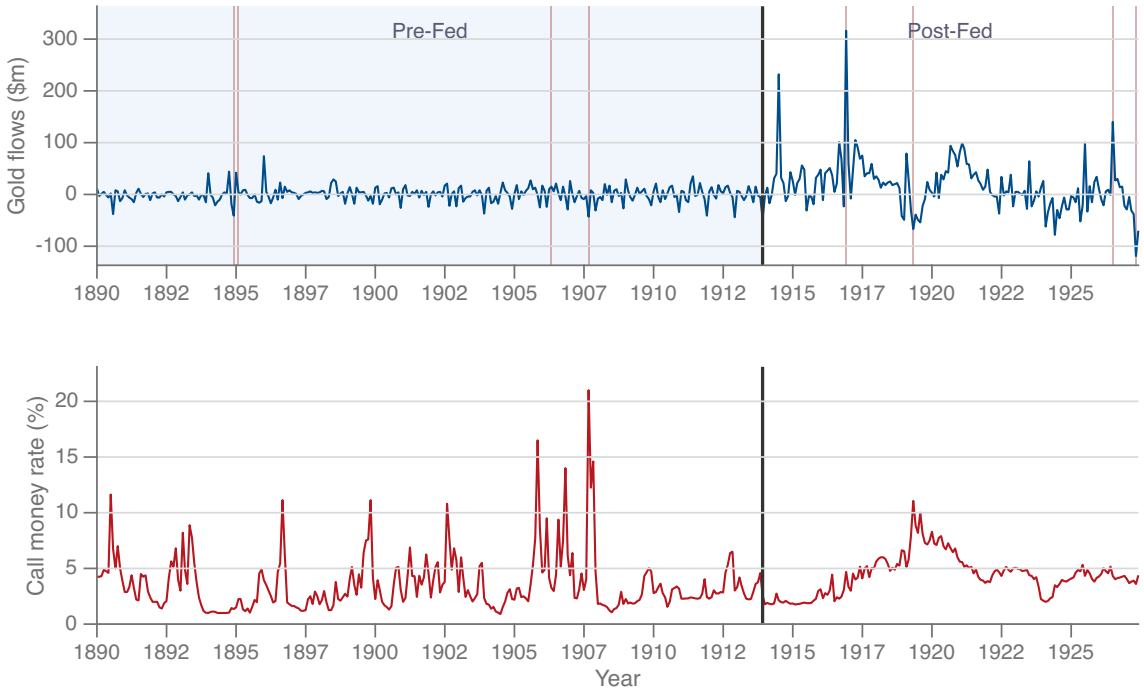
This paper asks one question. Before there was a central bank to conduct monetary policy, gold did the job. A gold inflow raised the monetary base and eased the money market, and a gold outflow tightened it, all without any decision by an authority. Our question is whether the founding of the Federal Reserve in 1913 changed that. Did the arrival of a central bank weaken the automatic link between gold movements and domestic monetary conditions, and if so, by how much and through which channels?

Gold has long stood at the center of the world economy. Its importance has faded since the world left the gold standard, but in the era we study the mere movement of gold could send ripples across markets and, more to our point, feed straight into the domestic money supply. Even so, few studies have looked closely at how gold flows moved the main economic aggregates during the gold standard, and fewer still have asked whether the founding of a central bank changed that link.

The reason the question is worth asking shows up in the raw data. Figure 1 plots the monthly change in the U.S. gold stock and the New York call money rate from 1890 to 1927, with the eight episodes we study marked and the 1913 Federal Reserve Act dividing the sample. Two things stand out. Gold movements did not get smaller after 1913. If anything the largest flows, driven by wartime Allied payments, came after the Fed. Yet the call money rate, which before 1913 lurched again and again into double digits and touched twenty-one percent during the Panic of 1907, becomes visibly calm afterward. Figure 2 makes the same point in one line. A rolling regression of the monthly change in the call rate on the gold flow is large and unstable before 1913 and then flattens toward zero. Something in the mapping from outside gold movements to the domestic money market seems to have changed, even as the gold movements themselves did not shrink. The task is to separate how much of that change reflects gold shocks in particular from how much reflects the institution that arrived in between.

The closest study to ours is Tallman and Moen (1998), who show that, in line with most of the anecdotal record, shocks to the stock of gold had lasting and significant effects on the economy during the National Banking Era, working through a short-run liquidity effect on interest rates. Between the end of their sample and the U.S. exit from gold, two institutional changes reshaped the setting. The Aldrich and Vreeland Act of 1908 let banks issue emergency currency during panics, a short-term fix meant to hold the system together until a fuller arrangement could be built. The Federal Reserve Act of 1913 then created a formal monetary authority with a lender of last resort, the power to vary reserve

Figure 1: Gold flows and the money market, 1890 to 1927



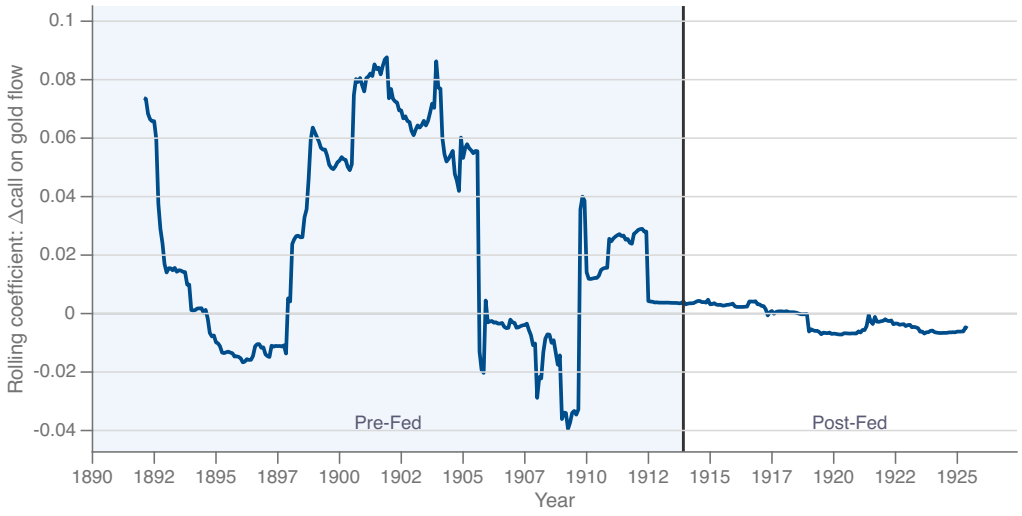
Notes. The top panel plots the monthly change in the U.S. gold stock in millions of dollars. The bottom panel plots the New York call money rate in percent. Thin vertical lines mark the eight episodes of Table 1. The heavy vertical line marks the Federal Reserve Act of December 1913, and shading marks the pre-Fed period. Gold flows do not shrink after 1913, but the call money rate becomes far less volatile.

requirements, the ability to move the discount rate apart from gold, and, in time, the capacity to buy and sell in the open market. Each of these tools makes it possible to offset a gold movement rather than let it run straight into the monetary base.

Open questions remain. Tallman and Moen (1998) find a short-run liquidity effect and no output effect for the National Banking Era as a whole, but we do not know whether the same holds in the narrower and more turbulent window around the Panic of 1907, or whether it survives the founding of the Fed. Answering this improves our grasp of monetary policy before there was a central bank to run it, and it speaks to the speed of transmission. Did the aggregates respond more slowly once guarantees were in place, or did expectations adjust faster and move them more quickly? Tracing how the power of gold shocks changed across one of the great turning points in American monetary history is, we argue, a real gap in the literature.

To answer these questions we pair a close historical account with a modern structural vector autoregression. Rather than lean on a continuous proxy for gold shocks, we use the

Figure 2: The sensitivity of the money market to gold flows over time



Notes. The line is the slope from a rolling regression of the monthly change in the call money rate on the gold flow, using a centered forty-eight-month window. It is large and unstable before 1913 and settles near zero after the Fed. This is a reduced-form correlation and its sign is not identified. The structural work in the body of the paper supplies the sign and the reading.

exogeneity of well-documented events through narrative sign restrictions in the manner of Antolín-Díaz and Rubio-Ramírez (2018). These restrictions force the model to respect a story for which many studies and primary accounts have supplied evidence, and they let us name a gold shock at specific dates without having to defend the relevance or exogeneity of a whole proxy series.

Our paper sits at the meeting point of three literatures. The first is the work on gold flows and the money market under the gold standard, from Neuburger and Stokes (1979) on the two-way tie between gold and interest rates to Bazot et al. (2016) on how an active central bank could offset gold movements through its own balance sheet. The second is the study of the U.S. National Banking Era, above all Tallman and Moen (1998), who document a liquidity effect of gold on interest rates using data that stop in 1909. We extend that reading in two ways that the earlier literature leaves open. We carry the analysis through the founding of the Fed and into the 1920s, so that the same shock can be measured on both sides of the institutional break, and we study gold outflows alongside inflows, since offsetting should weaken transmission in both directions and the outflow episodes are then a test of the same mechanism rather than a special case. The third is the literature on narrative and sign-restricted identification, from Uhlig (2005) and Arias et al. (2018) to the narrative restrictions of Antolín-Díaz and Rubio-Ramírez (2018) and the shrinkage prior of Giannone

et al. (2015), which together let us name a gold shock at documented dates without a proxy series. We keep the identification light, imposing only the three impact signs and two impact zeros implied by the model of Section 3 and leaving the rest of the response path free. The model's further predictions, that the liquidity effect persists and that output and prices rise with a lag, then become predictions we test against the estimates rather than restrictions we assume.

Our results point to a large change in the transmission of unexpected gold flows across the founding of the Federal Reserve. Transmission before the Fed is stronger, and the change is sharpest in the money market. When we let the dynamics differ across periods, the impact of a gold inflow on the call rate is about nine times larger before 1913, with credible bands that do not overlap, and the post-Fed response cannot be told apart from zero. We attribute this to the new ability of the Federal Reserve to offset gold movements and to lean on other monetary tools, and we set out the mechanism in a compact model in which two policy parameters govern how much of a gold flow is offset.

The paper proceeds as follows. Section 2 reviews the stories behind the eight episodes and then places the exercise in the wider literature. Section 3 develops the model that motivates the restrictions. Section 4 lays out the framework and the identification. Section 5 presents the results. Section 6 concludes.

2 Gold, Panics, and Central Banking

Our identification rests on the claim that at eight specific dates the sign and the dominance of the gold shock are not in serious historical doubt. That claim is only as strong as the stories behind it, so we start with the episodes themselves. We then place the exercise in the wider literature on gold flows, panics, and the move to central banking.

2.1 The eight episodes

The 1893 to 1895 Treasury reserve crisis and the Morgan-Belmont syndicate. The first pre-Fed inflow and outflow are two sides of one story. The Panic of 1893 and the drain of gold from the Treasury that followed, driven by doubts about whether the country would keep the gold standard under the Sherman Silver Purchase Act, pushed the Treasury's free gold reserve below the closely watched hundred-million-dollar mark and toward exhaustion by the winter of 1894 (Friedman and Schwartz, 1963; Timberlake, 1993). December 1894 records one of the largest single-month outflows in the pre-Fed sample. The crisis broke in February 1895 when a syndicate led by J.P. Morgan and August Belmont

agreed to supply the Treasury with gold, much of it drawn from abroad, in return for bonds. Many read the episode as private finance standing in for a missing lender of last resort (Friedman and Schwartz, 1963). We use the December 1894 drain as a pre-Fed outflow and the syndicate's inflow as a pre-Fed inflow.

The San Francisco earthquake, 1906. The April 1906 earthquake and fire destroyed much of the capital stock of a major American city and set off large insurance payments, a good share of them owed by British insurers. Odell and Weidenmier (2004) show that the resulting shipment of gold from London to the United States over the following months tightened conditions in Britain and helped set in motion the chain of events that ended in the Panic of 1907. For us the episode is close to ideal. It is an outside, insurance-driven inflow of gold whose source lies in a natural disaster rather than in U.S. financial conditions. We use the spring 1906 inflow as a pre-Fed inflow.

The Panic of 1907. The Panic of 1907 is the central pre-Fed episode. It began with the failed corner of United Copper and the runs on the Knickerbocker Trust and other trust companies, and it produced a scramble for cash, a suspension of convertibility, and a jump in the call rate to twenty-one percent (Sprague, 1910; Andrew, 1908; Moen and Tallman, 1992; Wicker, 2000). Two gold movements bracket the panic. As the crisis worsened in the autumn, gold drained from New York, and we use the September 1907 drain as a pre-Fed outflow. Rodgers and Payne (2014) then show that the Bank of France's November move to release gold in exchange for discounted commercial paper signaled an outside increase in the supply of gold to the U.S. banking system and helped end the panic. One caution is in order here. The Bank of France was also shipping gold to the United States around the same time, so the autumn of 1907 saw both a domestic drain and an outside inflow within a short span. We date the outflow shock to September, ahead of the November French shipments, so that the two do not fall in the same window, but the reader should keep the overlap in mind. Even so, this is the clearest case in our sample of a money market under acute stress, where a given gold movement moved the rate especially far.

Allied wartime payments, 1916. Our first post-Fed episode is the flood of gold that came with Allied purchases of U.S. goods during the First World War. By 1916 the belligerents were paying for enormous orders of munitions, food, and raw materials, and gold poured into the United States on a scale with no match in our sample. December 1916 records the single largest monthly inflow (Friedman and Schwartz, 1963; Meltzer, 2003). Crucially, this inflow came under a working Federal Reserve that could, and did, respond. We use it as a

post-Fed inflow and treat it as the sharpest test of the offsetting story, precisely because the raw shock was so large.

The 1919 gold-export embargo lift. During the war the United States held back gold exports. The embargo came off in June 1919 (Meltzer, 2003; Eichengreen, 1992). Its removal was followed by sizable outflows as gold was again free to move to settle foreign accounts. May 1919 records one of the larger outflows of the post-Fed period, and the episode is a clean post-Fed outflow driven by a clear policy change rather than by domestic conditions.

The 1926 gold-standard influx and the 1927 Strong-Norman episode. The last two episodes belong to the restored gold standard of the mid-1920s. With sterling back on gold from 1925 and the system rebuilt around a gold-exchange standard, gold moved in and out of New York in large amounts as the system adjusted (Eichengreen, 1992; Clarke, 1967; Chandler, 1958). We use the July 1926 influx as a post-Fed inflow. For the post-Fed outflow we use May 1927, the period around the understanding between Benjamin Strong and Montagu Norman under which the Federal Reserve eased policy in part to relieve pressure on sterling and draw gold toward London. May 1927 records the single largest monthly outflow in the whole sample (Chandler, 1958; Clarke, 1967). That the Fed was by this point using domestic policy to manage the outside gold account is itself the thing this paper is about.

2.2 Related literature

Two strands of work frame what follows. The first asks whether the founding of the Fed changed the behavior of the American money market at all. Miron (1986) shows that the sharp seasonal swings in nominal interest rates that characterized the National Banking Era largely disappear after 1914, which he reads as the Fed smoothing seasonal money demand, and Canova (1991) compares the incidence of financial panics before and after the Fed. Our question is the same in spirit but about a different forcing variable, namely gold rather than the seasons. The second strand asks whether central banks could insulate domestic conditions from international flows. Bazot et al. (2022) show that central banks under the classical gold standard routinely sterilized external flows, so that the textbook automatic adjustment mechanism operated far less mechanically than the price specie flow story implies.

The two-way tie between gold movements and interest rates is the oldest theme in this literature. Neuburger and Stokes (1979) study monthly data for Britain, France, and Germany in the late nineteenth and early twentieth centuries and ask both how gold flows triggered rate changes and how those changes shaped later gold movements. They find that in Britain

gold flows led to rate moves that in turn shaped future flows, while the French and German patterns were more mixed, and, most useful for us, that the strength of the tie depended on how actively the monetary authority responded. The way an active authority could sever that tie is shown directly by Bazot et al. (2016), who find that the Bank of France held its discount rate steady under the classical gold standard by adjusting its own portfolio rather than the rate, offsetting outside gold flows through domestic credit. Their result is the key precursor to this paper. An institution with a balance sheet and the will to use it can break the mechanical link between outside gold movements and domestic conditions. Questions of gold-point arbitrage and the bounds within which the exchange rate could move before shipments paid off are treated in full by Officer (1996).

For the United States, Tallman and Moen (1998) are the direct forerunner of this study. They document a short-run liquidity effect of gold on interest rates during the National Banking Era, with little effect on output, using monthly data from January 1890 to December 1909. We add to their work in three ways. We focus on identified episodes rather than the era as a whole, we carry the analysis through the founding of the Fed and into the 1920s, and we add gold outflows to the inflow-only design. The instability of the National Banking system that made gold flows so potent, from the inelastic currency to the pyramiding of reserves in New York and the recurring autumn scramble for cash, is set out by Sprague (1910), Timberlake (1993), and Wicker (2000), and the special role of the trust companies in 1907 by Moen and Tallman (1992). Rodgers and Payne (2014) give the cleanest case in the literature of an identified, plausibly outside gold shock changing U.S. asset-market expectations, which is just the kind of variation our narrative restrictions use.

The standard account stresses the limited activism of the early Federal Reserve and its failure to head off the 1920 to 1921 downturn (Meltzer, 2003). A more careful reading, and one our results speak to, is that even a passive Fed could shield the domestic money market from outside gold flows through tools the National Banking Era lacked. The wartime and early postwar episodes are covered by Meltzer (2003) and, for the remarkable market closure of 1914, by Silber (2007). The personalities and the international cooperation of the mid-1920s, including the Strong-Norman ties, are the subject of Chandler (1958) and Clarke (1967). The broader reading of the gold standard as a commitment device and of its fragility between the wars is developed by Bordo and Kydland (1995) and Eichengreen (1992), and the monetary history around all of these events remains that of Friedman and Schwartz (1963).

On method we build on the sign-restriction approach of Uhlig (2005), the formal treatment of inference under sign and zero restrictions in Arias et al. (2018), which in turn builds on Rubio-Ramírez et al. (2010), and the narrative sign restrictions of Antolín-Díaz and

Rubio-Ramírez (2018), which let information about the sign of a shock or the main driver of a variable at given dates sharpen an otherwise set-identified model. Our use of the shrinkage prior of Giannone et al. (2015) places the exercise in the modern Bayesian VAR tradition and, as we note below, is what makes the impulse responses smooth enough to read at a monthly frequency. Relative to the reduced-form and proxy-based work on gold shocks, the narrative approach lets us use exactly the episodes for which the record is strongest, while sidestepping the measurement error and exogeneity worries that come with a continuous proxy.

3 A Simple Model of Gold Flow Transmission

Before turning to the estimation, we set out the economics we have in mind. We build a compact monthly model of the gold standard that links an outside gold flow to high-powered money, the call money rate, activity, prices, the exchange rate, and equities. The model is kept simple on purpose. It uses accounting identities and linear behavioral relations so that it maps straight into the six series in y_t, it keeps the high-frequency mechanism stressed by contemporaries and by the literature (Friedman and Schwartz, 1963; Eichengreen, 1992; Bordo and Kydland, 1995; Tallman and Moen, 1998). The model plays two roles. It delivers the three impact signs and two impact zeros that we impose in Section 4, which discipline the identification, and it delivers a set of short- and medium-run predictions that we deliberately leave free and later test against the estimates. Its purpose is not to be estimated but to make precise the sense in which transmission should differ across periods.

3.1 The monetary base, cover, and offsetting

Let G_t be the nominal gold stock and M_t the monetary base. Under the gold standard the base is tied to gold through a cover relation,

$$M_t = \mu_t G_t, \qquad \mu_t \equiv \theta_t^{-1} \upsilon_t \in (0, \infty), \tag{1}$$

where θ_t is the gold cover ratio and υ_t stands for reserve preferences and the note-to-deposit mix. Gold flows follow

$$\Delta G_t = \rho_g \Delta G_{t-1} + \varepsilon_t^g, \qquad \rho_g \in (0,1), \qquad \mathbb{E}_{t-1}\varepsilon_t^g = 0, \tag{2}$$

so the monthly base innovation is $\Delta M_t = \mu_t \varepsilon_t^g + G_{t-1}\Delta\mu_t$. The feature that sets the model apart is that the monetary authority, when it exists, may offset a gold flow and smooth the

call rate,

$$\Delta\mu_t = -\phi_\mu \varepsilon_t^g - \psi_r(r_t - \bar{r}) + \eta_t, \qquad \phi_\mu, \psi_r \geq 0, \tag{3}$$

where ϕ_μ captures state-contingent offsetting and ψ_r captures rate-smoothing open-market activity. Before the Federal Reserve the market ran without such controls and we take $(\phi_\mu, \psi_r) \approx (0, 0)$. With a central bank in place, both can be strictly positive. Putting (3) into the base identity, the effect of a gold shock on the base is

$$\frac{\partial \Delta M_t}{\partial \varepsilon_t^g} = \mu_t - G_{t-1}\phi_\mu - G_{t-1}\psi_r \frac{\partial r_t}{\partial \varepsilon_t^g}. \tag{4}$$

3.2 Money demand and the liquidity effect

Money demand is $M_t = P_t L(y_t, r_t; \xi_t^m)$, which in log-linear form near a steady state is $m_t - p_t = \lambda y_t - \sigma r_t + \xi_t^m$ with $\lambda, \sigma > 0$. Combining this with (4) gives the liquidity effect.

Lemma 1 (Liquidity effect)**.** *Under* (1) *through* (3) *and money demand as above, the impact response of the call money rate to a gold shock is*

$$\frac{\partial r_t}{\partial \varepsilon_t^g} = -\frac{\mu_t - G_{t-1}\phi_\mu}{\sigma + G_{t-1}\psi_r}. \tag{5}$$

It is strictly negative whenever $\mu_t > G_{t-1}\phi_\mu$, and its size falls in both ϕ_μ and ψ_r.

Proof. Holding activity and the money-demand shock fixed within the month, differentiate the market-clearing condition $m_t - p_t = \lambda y_t - \sigma r_t + \xi_t^m$ with respect to ε_t^g, using $m_t = \log M_t$ and (4). Solving for $\partial r_t / \partial \varepsilon_t^g$ gives (5). The numerator is positive under $\mu_t > G_{t-1}\phi_\mu$ and the denominator is positive, so the derivative is negative. Differentiating (5) with respect to ϕ_μ and ψ_r shows that each raises it toward zero, so the size falls. □

Offsetting through ϕ_μ and rate smoothing through ψ_r both shrink the liquidity effect. If $\phi_\mu \geq \mu_t / G_{t-1}$ the effect can vanish or even flip. The sign restriction (12), which we impose only for the identified gold shock, rules out the flip within our episodes while leaving the size free to shrink.

3.3 Timing, predetermined output and prices

The model is monthly, and at that frequency it is not plausible that real output and posted prices respond within the same month to a gold shock. Output in month t reflects input orders, labor contracts, and mill schedules set in month $t - 1$, and a large share of prices

in this era were list or contract prices reset at set intervals. We therefore make industrial production predetermined,

$$\Delta IP_t = \chi_y \Delta y_{t-1} - \chi_r (r_{t-1} - r^*_{t-1}) + \nu^m_t, \qquad \chi_y, \chi_r > 0, \tag{6}$$

and prices set one period ahead in the manner of Fischer and Taylor,

$$p_t = \mathbb{E}_{t-1}\big[p^\star_t\big], \qquad p^\star_t = p_{t-1} + \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t + u_t, \qquad \beta \in (0,1),\ \kappa > 0. \tag{7}$$

Iterating the Euler equation for demand, $y_t = \mathbb{E}_t y_{t+1} - \phi(r_t - \mathbb{E}_t \pi_{t+1} - r^*_t) + \varepsilon^y_t$, forward shows that the expectation of future offsetting and smoothing, meaning higher ϕ_μ and ψ_r, raises the expected path of real rates after a gold shock and so damps the output response at every horizon.

3.4 Exchange rate and equities

Inside the gold points, uncovered interest parity gives a small near-impact move in the exchange rate, $\Delta s_t = \gamma(r_t - r^*_t) + \nu_t$ with $\gamma > 0$, so the lower U.S. rate after an inflow brings a dollar depreciation that fades as the rate gap closes. Shipments bind, and feed back into G_t, only when the exchange rate reaches a gold point. A linear dividend-discount relation, $\Delta q_t = \zeta_y \Delta IP_t - \zeta_r (r_t - r^*_t) + \xi^q_t$ with $\zeta_y, \zeta_r > 0$, closes the model. Because ΔIP_t is predetermined by (6), the impact equity response comes entirely through the discount-rate channel, a sharper prediction than a same-month formulation gives.

3.5 Comparative statics across periods

The key parameters are ϕ_μ and ψ_r. Before the Fed $(\phi_\mu, \psi_r) \approx (0, 0)$, so gold inflows map almost one-for-one into base money. With the Fed in place, $\phi_\mu > 0$ and $\psi_r \geq 0$ become feasible.

Proposition 1 (Comparison across periods)**.** *Under (1) through (7), the exchange-rate and equity relations above, and $\mu_t > G_{t-1}\phi_\mu$, a positive gold inflow shock $\varepsilon^g_t > 0$ implies the following.*

(i) At impact, $\partial \Delta G_t / \partial \varepsilon^g_t > 0$, $\partial r_t / \partial \varepsilon^g_t < 0$, $\partial \Delta q_t / \partial \varepsilon^g_t > 0$, and $\partial \Delta IP_t / \partial \varepsilon^g_t = \partial \pi_t / \partial \varepsilon^g_t = 0$.

(ii) In the short run, $h = 1, 2, 3$, $\partial r_{t+h} / \partial \varepsilon^g_t < 0$, $\partial \Delta q_{t+h} / \partial \varepsilon^g_t > 0$, and $\partial \Delta s_{t+h} / \partial \varepsilon^g_t < 0$.

(iii) In the medium run, $h = 6, 12$, $\partial \pi_{t+h} / \partial \varepsilon^g_t > 0$ and $\partial \Delta IP_{t+h} / \partial \varepsilon^g_t > 0$.

(iv) Every non-zero response in (i) through (iii) is weakly smaller in size in (ϕ_μ, ψ_r), strictly so whenever either is positive. Transmission is therefore strictly stronger before the Fed.

Proof. Parts (i) through (iii) follow by construction. The impact signs of ΔG, r, and Δq are immediate from (2), Lemma 1, and the discount-rate channel. The impact zeros for ΔIP and π follow from the predetermination in (6) and (7). Because the gold flow is persistent, (2) gives $\partial r_{t+h}/\partial \varepsilon_t^g = \rho_g^h \, \partial r_t/\partial \varepsilon_t^g < 0$ for $h \geq 0$, and uncovered interest parity carries the sign to Δs. The equity sign follows from the same rate path. The medium-run price and output signs follow from iterating (7) and (6) forward with the lasting fall in real rates. For part (iv), Lemma 1 shows $|\partial r_t/\partial \varepsilon_t^g|$ falls in (ϕ_μ, ψ_r). Since every other non-zero response is proportional to the rate path, either directly through Δs and the discount-rate term in Δq, or through the forward-iterated real rate in ΔIP and π, each inherits the same monotonicity, strictly whenever the matching derivative in Lemma 1 moves. □

Corollary 1 (The impact zeros carry no cross-period content)**.** *The impact restrictions $\partial \Delta IP_t/\partial \varepsilon_t^g = 0$ and $\partial \pi_t/\partial \varepsilon_t^g = 0$ hold in both periods and so say nothing about (ϕ_μ, ψ_r).*

Proof. Both derivatives are zero for any (ϕ_μ, ψ_r) by (6) and (7), which do not involve the policy parameters at horizon zero. So they cannot differ across periods. □

Corollary 1 shapes how we read the results. The impact zeros are timing restrictions, not period restrictions, so all cross-period content in Proposition 1 comes from the impact responses of r and Δq and from the whole $h \geq 1$ path. This is just why the comparison in Section 5.4 rests on separately estimated subsamples rather than on the impact responses alone.

By symmetry the same statements hold with every sign reversed for a gold outflow, $\varepsilon_t^g < 0$. An outflow raises the call rate, lowers equities and activity, and firms up the dollar, with sizes again weakly smaller in (ϕ_μ, ψ_r). This is what makes the four outflow episodes a fresh test of the same mechanism.

3.6 The predictions we take to the data

It is useful to collect the model's implications in one place, separating what we will impose from what we will test. Proposition 1(i) gives the impact restrictions, namely the signs on gold flows, the call rate, and equities, and the two zeros on output and prices. Those we impose. Parts (ii) and (iii) give a path,

$$\theta_{r,g}(h) < 0 \text{ and } \theta_{\Delta s,g}(h) < 0 \; (h = 1, 2, 3), \qquad \theta_{\Delta IP,g}(h) > 0 \; (h = 3, 6, 12), \qquad \theta_{\Delta p,g}(h) > 0 \; (h = 6, 12), \tag{8}$$

which we do *not* impose. Section 4 explains why, and Section 5 confronts (8) with the estimates.

4 Methodology

We estimate a Bayesian structural vector autoregression identified by sign, zero, and narrative restrictions in the spirit of Uhlig (2005), Arias et al. (2018), and Antolín-Díaz and Rubio-Ramírez (2018). Inference follows Arias et al. (2018), adjusted for the reweighting of Antolín-Díaz and Rubio-Ramírez (2018), and we use the shrinkage prior of Giannone et al. (2015). We first fix notation and the mapping from the structural to the reduced form, and then set out the identification, the narrative restrictions, and the prior.

4.1 Specification and notation

Let y_t denote the $n = 6$ vector of monthly variables

$$y_t = \left[\Delta IP_t,\ r_t^{\text{call}},\ \Delta q_t,\ \Delta s_t^{\text{USD/GBP}},\ \Delta p_t,\ \Delta G_t\right]',$$

where ΔIP_t is the monthly log change of the Miron and Romer industrial production index, r_t^{call} is the New York call money rate,[1] Δq_t is the monthly equity return, $\Delta s_t^{\text{USD/GBP}}$ is the monthly log change of the USD/GBP exchange rate, Δp_t is monthly inflation, and ΔG_t is the change in the U.S. gold stock in millions of dollars. One point on the call rate is worth stating plainly. It is constructed from *daily* New York call money quotations as in Fohlin (2025), not from the monthly averages carried in the NBER Macrohistory database. The distinction matters for episodes such as the Panic of 1907, when the rate moved violently within the month and a monthly average understates both the level and the timing of the strain. runs from 1890:01 to 1927:06, so $T = 450$.

The structural VAR is written as

$$y_t' A_0 = \sum_{l=1}^{p} y_{t-l}' A_l + c + \varepsilon_t', \qquad t = 1, \ldots, T, \tag{9}$$

where $A_l \in \mathbb{R}^{n \times n}$ for $l = 0, 1, \ldots, p$, A_0 has full rank, and c is a $1 \times n$ vector of constants. The $n \times 1$ vector of structural innovations ε_t is i.i.d. $N(0, I_n)$ given initial conditions. Collecting $A_+' = [A_1' \ \cdots \ A_p' \ c']$ and $x_t' = [y_{t-1}' \ \cdots \ y_{t-p}' \ 1]$, the compact form is $y_t' A_0 = x_t' A_+ + \varepsilon_t'$.

[1] Call money data collected and constructed as in Fohlin (2025).

Premultiplying by A_0^{-1} gives the reduced form

$$y_t' = x_t' B + u_t', \qquad B = A_+ A_0^{-1}, \qquad u_t' = \varepsilon_t' A_0^{-1}, \tag{10}$$

so that $\Sigma \equiv \mathbb{E}[u_t u_t'] = (A_0 A_0')^{-1}$. Letting Θ collect (A_0, A_+), the structural shocks come out as $\varepsilon_t'(\Theta) = y_t' A_0 - x_t' A_+$. Dynamic responses are summarized by the $n \times n$ impulse response matrices $\{\Theta_h\}_{h \geq 0}$, whose (k, j) element $\theta_{kj}(h)$ is the response of variable k at horizon h to a shock j at time t. We set $\Theta_0 = A_0^{-1}$ and build Θ_h from the reduced-form coefficients.

We report responses to a gold inflow of one standard deviation of the gold-flow series, about 33 million dollars, which is in the range of the larger episodes we study. Industrial production, equities, the exchange rate, and prices are shown in percent, the call money rate in percentage points, and gold flows in millions of dollars. This keeps the effects at a readable size and keeps the pre-Fed and post-Fed magnitudes on the same footing.

Information criteria on a common effective sample point to a short lag structure. The Akaike criterion selects $p = 3$, while the Hannan-Quinn and Schwarz criteria both select $p = 1$. This follows from the fact that five of the six series are growth rates or returns and are close to white noise. Since a first-order system cannot produce the hump-shaped medium-run dynamics that the model of Section 3 predicts and that the identification restricts, we set $p = 6$, one step above the Akaike choice and in line with monthly practice. Results at $p = 3$ are similar in sign and a little less persistent.

4.2 Identification by sign and zero restrictions

We name one structural shock the gold shock and leave the rest unrestricted, so the model is set-identified with respect to a single shock. Following Arias et al. (2018), we draw the orthogonal reduced form, keep draws whose gold column meets the restrictions, and reweight by the proper volume elements so that the reported posterior is the right one given the zero restrictions.

We keep the identification deliberately light and let it bind only at impact, in line with the liquidity view of Tallman and Moen (1998). We impose three impact sign restrictions,

$$\theta_{\Delta G, g}(0) > 0, \qquad \text{(gold inflow rises)} \tag{11}$$

$$\theta_{r, g}(0) < 0, \qquad \text{(liquidity lowers money-market rates)} \tag{12}$$

$$\theta_{\Delta q, g}(0) > 0, \qquad \text{(equities improve on easier credit)} \tag{13}$$

together with two zero restrictions that rule out a same-month response of real activity and

prices,

$$\theta_{\Delta IP,g}(0) = 0, \qquad \theta_{\Delta p,g}(0) = 0. \tag{14}$$

Section 3 derives (14) from stated timing assumptions rather than imposing it by fiat. That is the whole of the identification, three impact signs and two impact zeros on the gold column, and in estimation about 13 percent of candidate rotations meet it.

We stop here on purpose. The model makes further predictions, collected in (8), that the liquidity effect persists for a few months, that the dollar depreciates on impact, and that output and prices rise with a lag. We do not impose them. We treat them instead as predictions to test against the estimates. Imposing the whole response path would force every accepted draw into nearly the same shape, so the estimated responses would look alike across episodes and their credible bands would understate the true uncertainty. Leaving the path free lets the data speak to the model's dynamic predictions and gives a transparent measure of what the impact restrictions alone can pin down. As Section 5 shows, the freed responses still trace out the shapes in (8), so the model's dynamics hold in the data without being assumed.

4.3 Identification by narrative sign restrictions

We add narrative restrictions of the two kinds in Antolín-Díaz and Rubio-Ramírez (2018). The first fixes the sign of the structural shock at an event date. The second makes the gold shock the main driver of a chosen series over an episode window. For the inflow and outflow episodes we impose, respectively,

$$\varepsilon_{g,t^\star} \geq 0 \quad \text{(inflow)}, \qquad \varepsilon_{g,t^\star} < 0 \quad \text{(outflow)}, \tag{15}$$

and, over the window $[t^\star, t^{\star\star}]$,

$$\left|H_{\Delta G,g,t^\star,t^{\star\star}}\right| - \max_{j \neq g} \left|H_{\Delta G,j,t^\star,t^{\star\star}}\right| > 0, \tag{16}$$

so that the gold shock is the single largest driver of the observed move in gold flows over the episode. The strength of this approach is that it uses the narrative without a proxy. We write the historical account straight into the structural parameters and avoid the measurement error and exogeneity concerns that come with a continuous instrument. Draws that meet (15) and (16) are kept and reweighted by importance sampling, following Antolín-Díaz and Rubio-Ramírez (2018), so that conditioning on the narrative does not distort the posterior.

4.4 Prior and computation

We use the shrinkage prior of Giannone et al. (2015), with the Minnesota, sum-of-coefficients, and single-unit-root parts, and with the hyperparameters chosen by maximizing the marginal likelihood. Shrinkage matters here because the system is richly parameterized. With $n = 6$ variables and $p = 6$ lags each equation has thirty-six coefficients, and an unregularized estimator would fit high-frequency features of the lag structure that carry no economic content and translate into jagged impulse responses. The shrinkage prior pulls the lag structure toward a random walk and delivers smooth responses, leaving their sign and broad shape to the identification and the data. We draw the posterior with the algorithm of Arias et al. (2018). We sample (B, Σ) from the Normal-Inverse-Wishart posterior, draw an orthogonal rotation, check the sign and zero restrictions on the gold column, form the importance weight from the volume elements, and then check the narrative restrictions and reweight following Antolín-Díaz and Rubio-Ramírez (2018).

4.5 The eight episodes

Table 1 lists the episodes and the restrictions we impose. The design is balanced by construction. Four episodes come before the Federal Reserve Act and four after the opening of the Reserve Banks, and within each period two are gold inflows and two are gold outflows. All pre-Fed episodes fall on or before 1907 and all post-Fed episodes on or after 1916, so none sits near the 1913 to 1914 boundary. The stories behind each are in Section 2.1.

We estimate on three kinds of sample. The main impulse response of Section 5.1 uses the full 1890 to 1927 sample. The period comparison of Section 5.4 uses the pre-Fed and post-Fed subsamples. And each episode response is estimated on its own local window, about seven years on either side of the event and clamped to the sample and to the event's own period, so that the dynamics are allowed to differ from one episode to the next rather than being tied to one common set of coefficients. The windows are short enough that the episode bands are wide, which is the price of letting each event's own data determine its dynamics.

5 Results

We report the results in four steps. We begin with the identified gold-shock impulse response on the full sample, then look at the response conditioned on each episode, then at the historical decompositions around each episode, and last at a comparison of the two periods based on separately estimated subsamples. Throughout, responses are to a gold inflow of one standard deviation, about 33 million dollars, and outflow episodes are shown as outflows.

Table 1: Narrative episodes

Period	Episode	Direction	Shock month	Contribution window
Pre-Fed	Treasury Gold-Reserve Crisis	Outflow	Dec 1894	Nov to Dec 1894
Pre-Fed	Morgan-Belmont Syndicate	Inflow	Feb 1895	Jan to Apr 1895
Pre-Fed	San Francisco Earthquake	Inflow	May 1906	Apr to Jul 1906
Pre-Fed	Panic of 1907 (drain)	Outflow	Sep 1907	Aug to Sep 1907
Post-Fed	Allied Wartime Payments	Inflow	Dec 1916	Oct 1916 to Jan 1917
Post-Fed	WWI Gold-Export Embargo Lift	Outflow	May 1919	Apr to Jul 1919
Post-Fed	Restored Gold-Standard Influx	Inflow	Jul 1926	Jun to Aug 1926
Post-Fed	Strong-Norman Accord	Outflow	May 1927	Apr to Jun 1927

5.1 The gold-shock impulse response

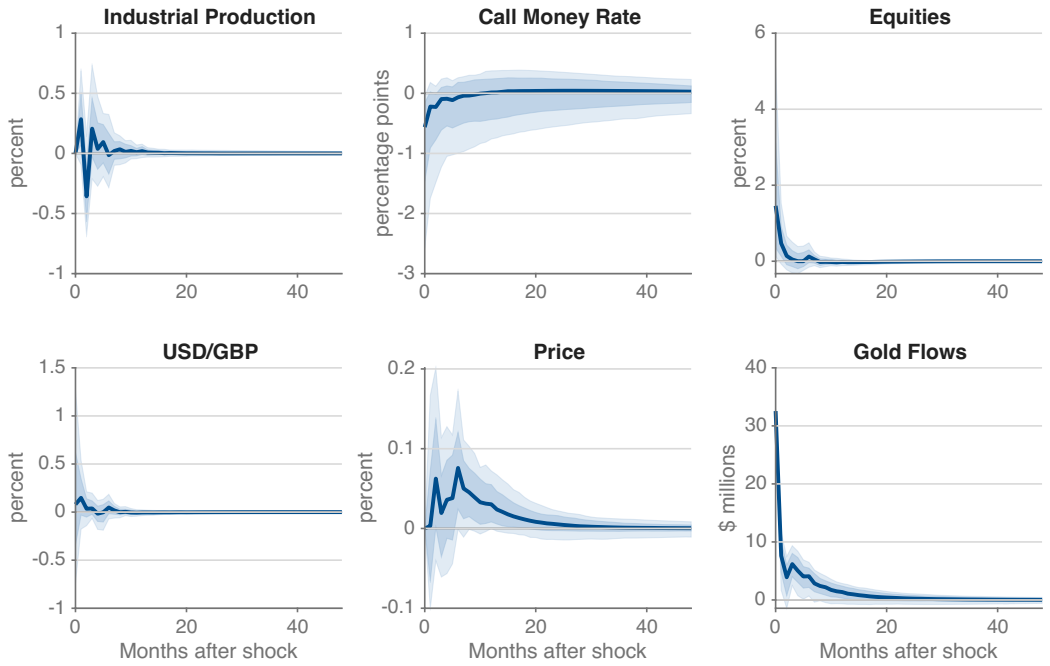
Figure 3 reports the impulse response to the identified gold shock on the full 1890 to 1927 sample. This is the central structural object of the paper, and because the identification binds only at impact, the response at every later horizon is left to the data. The figure is therefore also a test of the dynamic path the model predicts in (8).

The impact responses are the ones we restrict, and they come out as posed. A one standard deviation gold inflow lowers the call money rate by about 0.56 percentage points on impact, with a 68 percent band from about -1.5 to -0.2 points that stays below zero, and raises equities by about 1.5 percent. Output and prices are zero on impact by construction. What happens next is not imposed, and this is where the model earns its keep. The call rate stays below zero for several months before returning to trend, which is the liquidity effect of Lemma 1 appearing on its own. Industrial production rises to about 0.2 percent by three months, and the price level to about 0.08 percent by six months with a band that stays above zero, so the lagged pass-through built into the model in equations (6) and (7) shows up in the estimates without being assumed. The one prediction the data do not clearly deliver is the small dollar depreciation, where the median is close to zero and slightly the wrong sign, though the band is wide. Taken together, the freed responses trace out the path the model predicts in (8), which is stronger evidence for the mechanism than imposing that path would have been. The bands are correspondingly wide, an honest reflection of what the impact restrictions alone can pin down.

5.2 Impulse responses by episode

The full-sample response averages over the whole period. To let each event speak for itself we re-estimate the model on a local window around each one, as described in Section 4, so that each episode carries its own reduced-form dynamics. The eight responses, reported in

Figure 3: The gold-shock impulse response, full sample



Notes. Response to a gold inflow of one standard deviation on the full 1890 to 1927 sample, identified by the impact sign and zero restrictions of Section 4 with no narrative conditioning. Solid line is the posterior median, shaded regions are the 68 and 90 percent pointwise credible bands.

Appendix B, now differ across events, and they differ in a way that lines up with the paper's theme. Table 2 collects the impact responses.

The pattern is clear. Every pre-Fed episode shows a sizable liquidity effect on the call money rate, from about -0.6 points in the Morgan-Belmont inflow to about -2.6 points in the San Francisco earthquake, and in each case the 68 percent band lies below zero. The equity response is correspondingly large, between about three and five and a half percent. Every post-Fed episode, by contrast, is muted, with a call-rate response near -0.1 point and an equity response below one percent, and tight bands around those small numbers. The San Francisco and Panic of 1907 episodes carry the widest bands, since their windows are the noisiest, but even there the effect is clearly present and large. The post-Fed episodes look alike because their windows overlap and because there is little left to explain once the Fed offsets the flow, while the pre-Fed episodes differ from one another in both size and shape. Reading the episodes one at a time thus tells the same story as the period comparison below, now at the level of the individual event.

Notes. Posterior medians of the response to a one standard deviation gold inflow, each estimated on a local window around the event. Outflow episodes are normalized to a unit inflow for comparability. Full impulse

Table 2: Impact responses by episode, local-window estimates

Period	Episode	Call rate, impact (pp)	Equities, impact (%)
Pre-Fed	Treasury Gold-Reserve Crisis	−1.08	4.03
Pre-Fed	Morgan-Belmont Syndicate	−0.61	3.87
Pre-Fed	San Francisco Earthquake	−2.57	5.51
Pre-Fed	Panic of 1907	−1.29	3.01
Post-Fed	Allied Wartime Payments	−0.12	0.55
Post-Fed	Gold-Export Embargo Lift	−0.12	0.36
Post-Fed	Restored Gold-Standard Influx	−0.13	0.65
Post-Fed	Strong-Norman Accord	−0.16	0.65

responses with credible bands are in Appendix B.

5.3 Historical decompositions

Figures 4 and 5 report historical decompositions for two representative episodes. The other six are in Appendix C. Each panel plots the observed deviation from the deterministic baseline against the gold-shock contribution, under both the sign-only and the narrative posteriors, over a window of one year on either side of the episode. The shaded band marks the window over which the contribution restriction (16) binds.

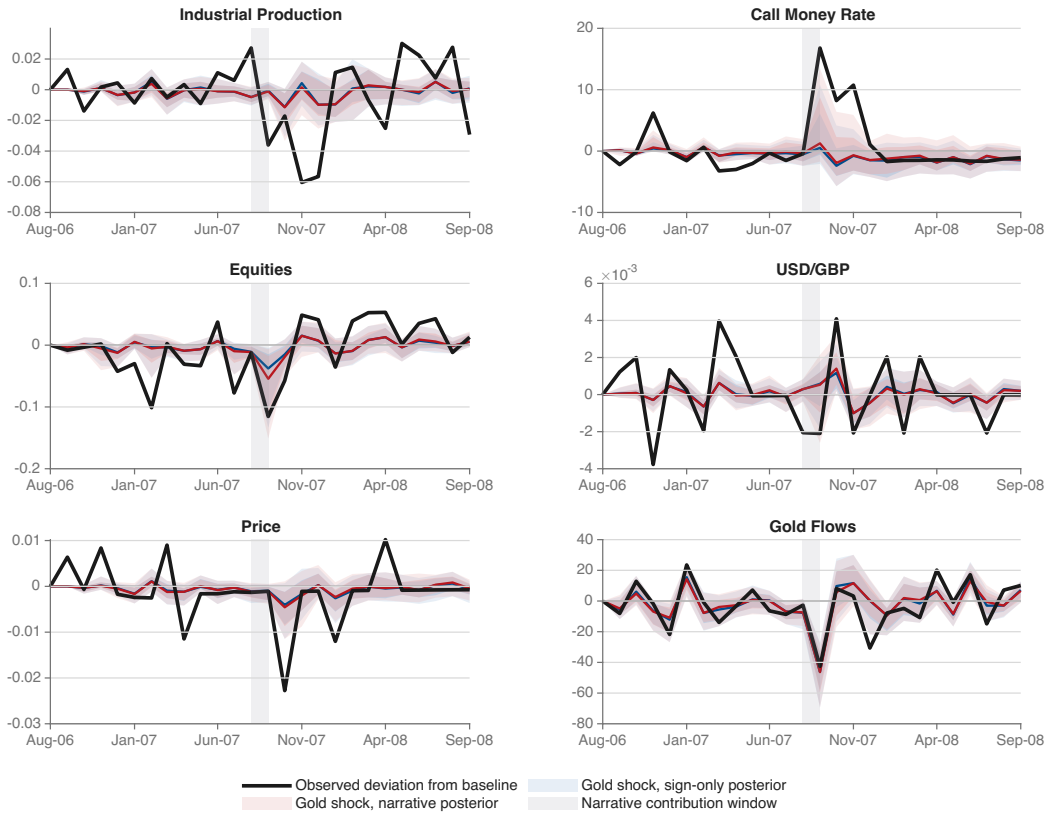
The Panic of 1907 decomposition is telling. The gold-flow panel shows an observed September 1907 drain of about forty-three million dollars, essentially all of which the narrative posterior assigns to the gold shock, at about forty-six million. The gold shock, in other words, accounts for the outflow itself. The call money rate is the more interesting panel. The observed rate spiked by nearly seventeen points that month, on its way to twenty-one percent, and the gold shock contributes about one and a quarter points of that, or roughly seven percent. The rate spike was overwhelmingly the panic itself, the runs and the scramble for cash, not the gold movement. So the gold shock drove the drain and nudged the rate in the same upward direction, but the money-market convulsion of 1907 was mostly other forces.

The Allied wartime payments decomposition tells a different and, for us, more useful story. The observed December 1916 inflow was about three hundred million dollars, of which the gold shock accounts for roughly two thirds. Its implied contribution to the call money rate is negative, about two-thirds of a point of easing, yet the observed rate that month rose by about two and a half points. The easing push from an enormous gold inflow did not show up in the money market. It was more than offset by other forces.

This is the mechanism in a single picture, though it needs care. The decomposition does

not label the offsetting force as policy. It assigns it to the residual shocks, which in 1916 and 1917 include war finance and a fast-rising demand for credit. What the figures do establish is that in the pre-Fed episodes the gold shock and the observed money-market move point the same way, while in the post-Fed episodes a gold-driven easing push is absorbed without moving the observed rate. Whether the force doing the absorbing was Federal Reserve offsetting or wartime credit demand cannot be settled by the decomposition alone.

Figure 4: Historical decomposition, Panic of 1907

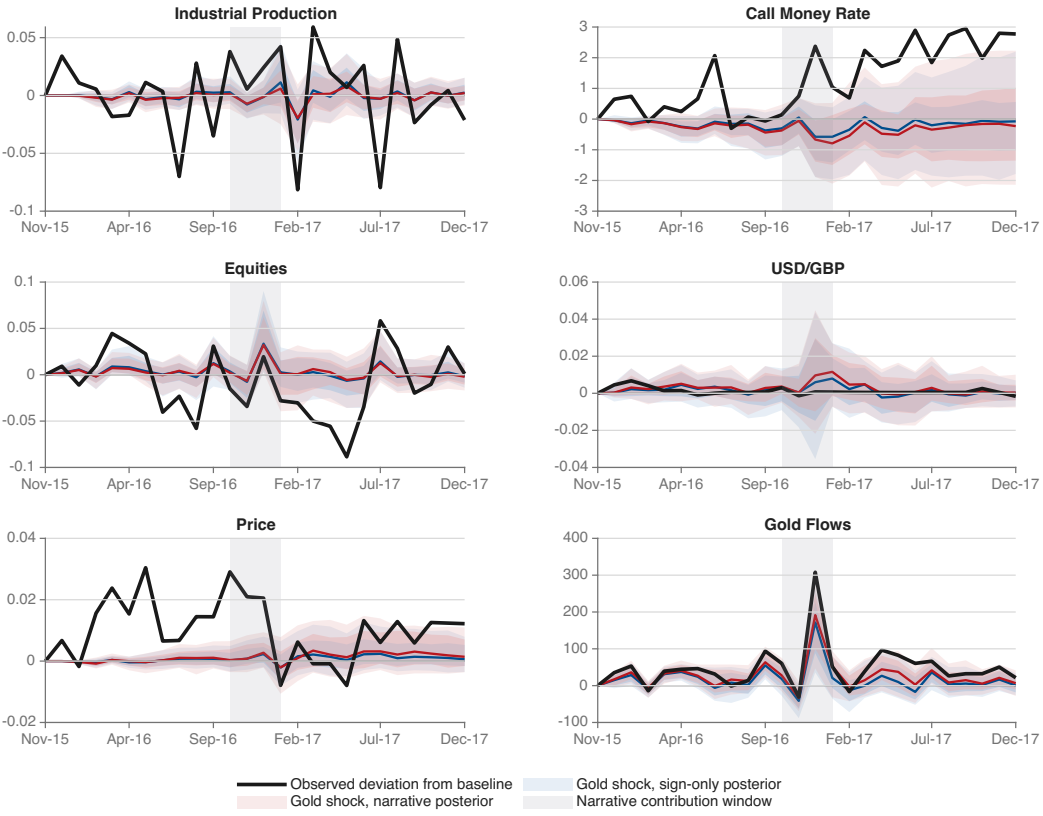


Notes. In each panel the black line is the observed deviation from the deterministic baseline, the blue band is the gold-shock contribution under the sign-only posterior, and the red band is the contribution under the narrative posterior, each with 68 and 90 percent bands. Shading marks the contribution window.

5.4 Comparison across periods

The episode results are conditioned on one common set of dynamics. To let the dynamics themselves differ, we re-estimate the model separately on the pre-Fed subsample, 1890:01 to 1913:12, and the post-Fed subsample, 1914:01 to 1927:06, identifying the gold shock by the

Figure 5: Historical decomposition, Allied wartime payments (December 1916)



Notes. See the notes to Figure 4.

same sign and zero restrictions but with no narrative conditioning, and scaling each draw to a one standard deviation gold inflow. Figure 6 and Table 3 report the result.

The gap is far sharper than the full-sample response can show. In the pre-Fed sample a one standard deviation gold inflow lowers the call money rate by about 1.0 percentage point on impact. In the post-Fed sample the matching response is about 0.1 point, roughly nine times smaller, and stays near zero at every horizon. The two 68 percent bands do not overlap. The pre-Fed band runs from about -3.3 to -0.4 points and the post-Fed band from about -0.3 to nearly zero. The equity and output responses show the same pattern, about five- and eight-fold, while the price response is essentially the same in the two periods. The exchange-rate response is, if anything, larger after the Fed, which fits a period in which outside adjustment runs through the exchange rate rather than through domestic rates.

Notes. Posterior medians, with 68 percent bands in brackets for the impact call-rate response. Each subsample uses 2000 accepted draws under the impact sign and zero restrictions of Section 4, with no narrative conditioning.

Table 3: Comparison across periods, response to a one standard deviation gold inflow

	Pre-Fed (1890 to 1913)	Post-Fed (1914 to 1927)	Ratio
Call rate, impact (pp)	−1.02	−0.11	8.9
	[−3.29, −0.42]	[−0.33, −0.04]	
Equities, impact (%)	2.67	0.49	5.4
Output, 3 months (%)	0.42	0.05	8.0
Prices, 6 months (%)	0.04	0.04	0.9

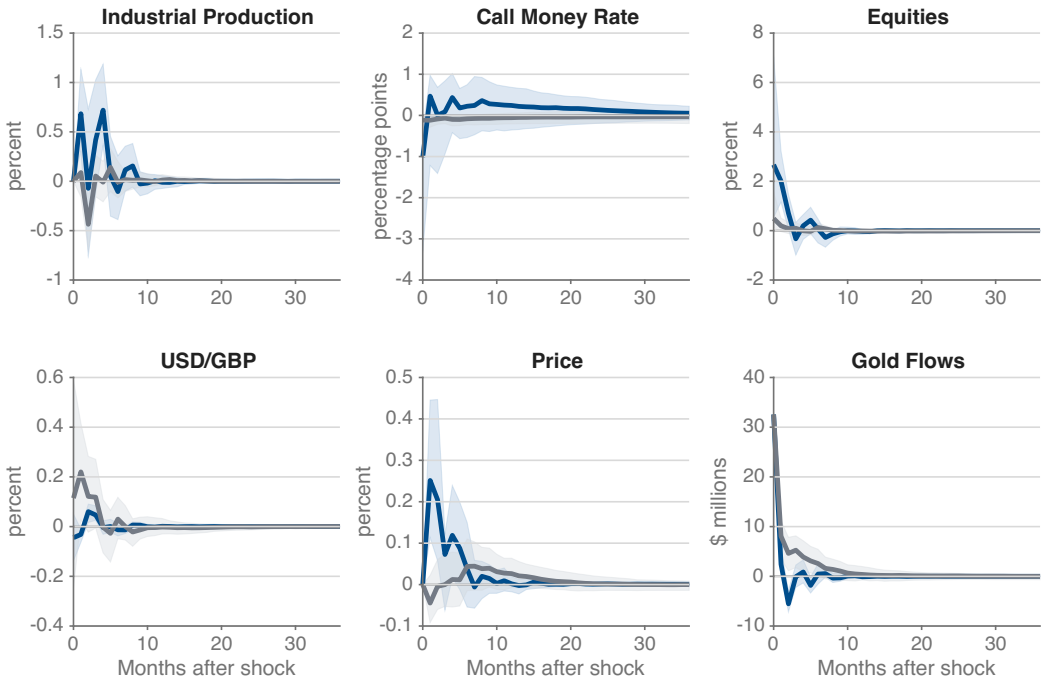
This is the paper's central piece of evidence, and it helps to be precise about what it does and does not show. It shows that the average gold shock, identified the same way in both samples, fed into the domestic money market before 1913 and mostly did not afterward. The gap is concentrated in the money market and in real activity. Prices, which respond only with a lag in either period, barely differ across the two, so the change is in how fast and how far gold moved the cost of credit, not in the eventual pass-through to the price level. Two cautions apply. The comparison does not by itself prove that offsetting was the mechanism, since the post-Fed sample also spans the war, the 1920 to 1921 deflation, and the restored gold standard, any of which could change the dynamics. And the pre-Fed size is loosely estimated, with a 68 percent band that spans a factor of about eight, so the ordering is what we lean on rather than the exact ratio.

The full-sample and episode responses are the complement to this comparison. They hold the sample fixed and so understate the gap, because a single reduced form cannot differ across periods, but they are not open to the objection that other post-1913 changes load onto the gold shock. The two exercises bracket the truth, and both point the same way. It is the direction and the robustness of the ordering, rather than either point estimate, that we take as the finding.

5.5 What the model says about the gap

It is worth pausing to read the results through the model of Section 3. The model ties the whole pattern to two policy parameters, ϕ_μ for offsetting and ψ_r for rate smoothing, both near zero before the Fed and both able to turn positive after. Three predictions of Proposition 1 line up with what we find. First, the liquidity effect of Lemma 1, $\partial r_t / \partial \varepsilon_t^g = -(\mu_t - G_{t-1}\phi_\mu)/(\sigma + G_{t-1}\psi_r)$, is exactly the object that shrinks toward zero as ϕ_μ and ψ_r rise. The call rate response is the one that collapses most across periods, from about -1.0 point to about -0.1, so if the data are speaking through this equation they say that ϕ_μ and ψ_r went from near zero to clearly positive. Second, the model makes the impact equity response a pure discount-rate effect once output is predetermined, so the equity gap should

Figure 6: Response to a gold inflow by period, separate subsample VARs



Notes. Response to a one standard deviation gold inflow, estimated separately on the pre-Fed sample (1890 to 1913, blue) and the post-Fed sample (1914 to 1927, grey), with 68 percent bands. The gold shock is identified by the sign and zero restrictions of Section 4 with no narrative conditioning.

track the call-rate gap rather than move on its own. It does, at about five-fold against the call rate's nine-fold. Third, the model predicts that prices, which move only with a lag and whose impact response carries no information about the policy parameters by Corollary 1, should show the smallest cross-period gap. The price gap is indeed the smallest, close to none at all, against roughly eight-fold for output. In short, the ranking of the gaps across variables, not just their common direction, is what the model predicts once offsetting turns on.

The ranking is the message rather than any single number. The variables that move through the call rate, the call rate itself and equities, show the largest gaps, real activity follows, and prices, which move only with a lag, show none. That is the signature of an authority that learned to break the tie between gold and the cost of credit while leaving the slow adjustment of the price level much as it was.

5.6 Local projection robustness

A natural worry is that the results are an artifact of the vector autoregression, which ties the response at every horizon to a single set of lag coefficients through its recursive structure. Local projections offer a check, because they estimate the response at each horizon from a separate regression and so impose no link across horizons (Jordà, 2005). We hold the identification fixed and let only the propagation go free. Let $\widehat{\varepsilon}_t^{\,g}$ denote the posterior-median structural gold shock recovered from the subsample VAR of Section 5.4, that is, the median across retained draws of the first element of $\varepsilon_t(\Theta) = y_t' A_0 - x_t' A_+$. For each outcome y in the system and each horizon $h = 0, 1, \ldots, 24$, and separately on the pre-Fed and post-Fed samples, we estimate the projection

$$y_{t+h} = a_h + b_h\, \widehat{\varepsilon}_t^{\,g} + \sum_{l=1}^{p} \boldsymbol{\gamma}_{h,l}'\, y_{t-l} + e_{t+h}, \qquad h = 0, 1, \ldots, 24, \tag{17}$$

where the lagged terms y_{t-l} absorb residual serial dependence and $p = 6$ matches the VAR. The impulse response at horizon h is the coefficient b_h, and the reported band is a 90 percent interval built from Newey-West standard errors with bandwidth $h + 4$, which allows for the moving-average dependence that overlapping h-step forecasts induce (Newey and West, 1987). Because $\widehat{\varepsilon}_t^{\,g}$ is a structural innovation and is therefore orthogonal to the past by construction, the estimate of b_h is consistent whether or not the controls in (17) are included. We scale b_h so that the impact response of gold flows equals one standard deviation of the gold-flow series, as in the rest of the paper.

Before turning to the figure, it is worth saying why the shock in (17) is the identified shock rather than a raw gold surprise. A reduced-form surprise, formed as the residual of the gold flow on its own past and the past of the other variables, does not work here. It carries the wrong sign for the call money rate, because under the gold standard a higher rate itself pulls gold in, so the raw correlation between a gold surprise and the rate is dominated by that reverse channel. Identification is what removes the reverse channel, so the local projection must be run on the identified shock to be a check on the structural result rather than on the reduced-form comovement.

Figure 7 reports the result. It matches the VAR closely. Before the Fed the call money rate falls by about 1.5 points on impact, with a Newey-West standard error near three-quarters of a point, so the effect is present and reasonably well estimated, and it returns toward zero over the first half year. After the Fed the same impact response is about -0.1 point and is not different from zero at any horizon. The equity, output, and exchange-rate responses repeat the pattern, larger and better determined before the Fed and flat afterward.

That an estimator which shares none of the VAR's cross-horizon structure delivers the same impact magnitudes and the same pre-Fed to post-Fed gap tells us the finding is not a product of the VAR's parametric form.

6 Conclusion

This paper studies how the transmission of gold-flow shocks, the de facto monetary policy tool of the age before central banking, changed across the founding of the Federal Reserve. Using a Bayesian structural VAR identified by sign, zero, and narrative restrictions, and conditioning on eight documented episodes from 1890 to 1927, we find that transmission was stronger before 1913 on every margin we measure, and that the gap was very uneven.

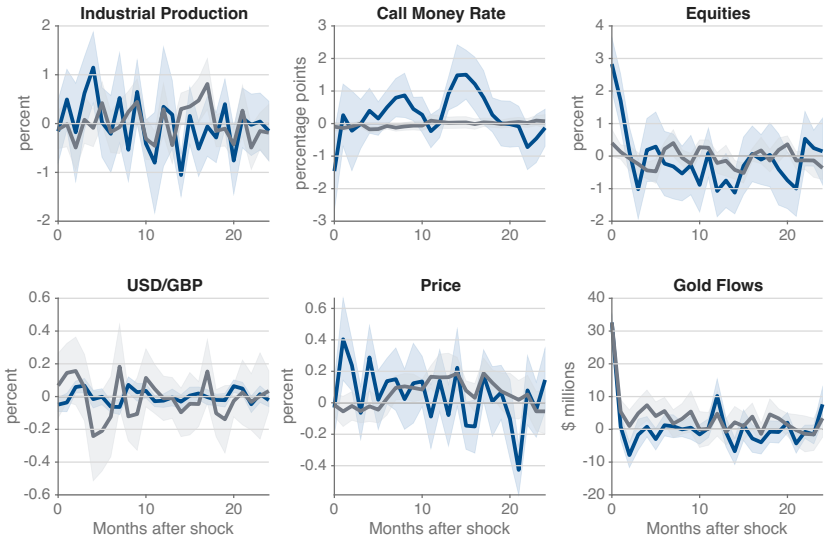
We identify the gold shock with light impact restrictions and leave the rest of the response free, and the estimated shock still lowers the call rate, lifts equities, and raises output and prices with a lag, so the model's dynamics come out of the data rather than being assumed. When we let the dynamics differ by estimating the two subsamples separately, the money-market gap is stark. The impact of a gold inflow on the call rate is about nine times larger before 1913, the two credible bands do not overlap, and the post-Fed response is close to zero at every horizon. The gap is concentrated in the money market and in real activity. Prices, which respond only with a lag in either period, barely differ across the two.

Our model explains this pattern through two policy parameters, ϕ_μ for state-contingent offsetting and ψ_r for rate smoothing. Both were near zero before 1913, so gold inflows mapped almost one-for-one into base money. With the Fed in place, positive values of these parameters shrink both the size and the persistence of gold shocks. Proposition 1 sets out the ranking, and the estimates respect it. Corollary 1 makes clear that the impact zeros we impose on output and prices are timing restrictions that carry no cross-period content, which is why the comparison rests on the subsample estimates.

Several limits deserve stress. The sample ends in 1927, before the Great Depression reshaped both gold-standard mechanics and Federal Reserve doctrine. The post-Fed subsample is short and spans several distinct settings, so the comparison cannot cleanly separate the arrival of the Fed from other changes at the same time. Narrative restrictions, while credible, limit the number of shocks we can study. And the pre-Fed subsample estimate carries a wide band, so it is the ordering across periods, which is clear and robust, rather than the exact ratio, that we rest on.

Future work might test the mechanism directly by bringing the Fed's discount-window and reserve-requirement operations to bear on the estimated offsetting parameters, compare the U.S. transition with other countries' moves from gold-standard automaticity to managed

Figure 7: Local projection responses to a gold inflow by period



Notes. Local projection responses to a one standard deviation gold inflow, estimated separately on the pre-Fed sample (1890 to 1913, blue) and the post-Fed sample (1914 to 1927, grey), with 90 percent Newey-West bands. The regressor is the posterior-median structural gold shock from the subsample VAR of Section 5.4, so the identification matches the main text and only the propagation is left free.

central banking, or let the transmission parameters drift through time rather than imposing a single break in 1913.

To close, gold kept its monetary power through the turn of the twentieth century, but mainly in the absence of an institution able to offset it. The founding of the Federal Reserve did more than add a lender of last resort. It rewired the link between outside shocks and the domestic money market. J.P. Morgan's line caught the pre-Fed world in which gold movements set monetary conditions directly. By the 1920s that link ran through policy rather than around it.

References

Andrew, A. P. (1908). Substitutes for cash in the panic of 1907. *Quarterly Journal of Economics*, 22(4), 497–516.

Antolín-Díaz, J. and Rubio-Ramírez, J. F. (2018). Narrative sign restrictions for SVARs. *American Economic Review*, 108(10), 2802–2829.

Arias, J. E., Rubio-Ramírez, J. F., and Waggoner, D. F. (2018). Inference based on structural vector autoregressions identified with sign and zero restrictions. *Econometrica*, 86(2), 685–720.

Bazot, G., Bordo, M. D., and Monnet, E. (2016). International shocks and the balance sheet of the Bank of France under the classical gold standard. *Explorations in Economic History*, 62, 87–107.

Bordo, M. D. and Kydland, F. E. (1995). The gold standard as a rule. *Explorations in Economic History*, 32(4), 423–464.

Bazot, G., Monnet, E., and Morys, M. (2022). Taming the global financial cycle. Central banks and the sterilization of capital flows in the first era of globalization. *Journal of Economic History*, 82(3), 801–839.

Canova, F. (1991). The sources of financial panic. Pre and post Fed evidence. *International Economic Review*, 32(3), 689–713.

Chandler, L. V. (1958). *Benjamin Strong, Central Banker*. Brookings Institution, Washington.

Clarke, S. V. O. (1967). *Central Bank Cooperation, 1924 to 1931*. Federal Reserve Bank of New York, New York.

Eichengreen, B. (1992). *Golden Fetters. The Gold Standard and the Great Depression, 1919 to 1939*. Oxford University Press, New York.

Fohlin, C. (2025). Daily call money rates and the New York money market, 1890 to 1930. Working paper, Emory University.

Friedman, M. and Schwartz, A. J. (1963). *A Monetary History of the United States, 1867 to 1960*. Princeton University Press, Princeton.

Giannone, D., Lenza, M., and Primiceri, G. E. (2015). Prior selection for vector autoregressions. *Review of Economics and Statistics*, 97(2), 436–451.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1), 161–182.

Meltzer, A. H. (2003). *A History of the Federal Reserve, Volume 1, 1913 to 1951*. University of Chicago Press, Chicago.

Miron, J. A. and Romer, C. D. (1990). A new monthly index of industrial production, 1884 to 1940. *Journal of Economic History*, 50(2), 321–337.

Miron, J. A. (1986). Financial panics, the seasonality of the nominal interest rate, and the founding of the Fed. *American Economic Review*, 76(1), 125–140.

Moen, J. and Tallman, E. W. (1992). The bank panic of 1907 and the role of trust companies. *Journal of Economic History*, 52(3), 611–630.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

Neuburger, H. M. and Stokes, H. H. (1979). The relationship between interest rates and gold flows under the gold standard. *Economica*, 46(183), 261–279.

Odell, K. A. and Weidenmier, M. D. (2004). Real shock, monetary aftershock. The 1906 San Francisco earthquake and the panic of 1907. *Journal of Economic History*, 64(4), 1002–1027.

Officer, L. H. (1996). *Between the Dollar-Sterling Gold Points*. Cambridge University Press, Cambridge.

Rodgers, M. T. and Payne, J. E. (2014). How the Bank of France changed U.S. equity expectations and ended the panic of 1907. *Journal of Economic History*, 74(2), 420–448.

Rubio-Ramírez, J. F., Waggoner, D. F., and Zha, T. (2010). Structural vector autoregressions. Theory of identification and algorithms for inference. *Review of Economic Studies*, 77(2), 665–696.

Silber, W. L. (2007). *When Washington Shut Down Wall Street*. Princeton University Press, Princeton.

Sprague, O. M. W. (1910). *History of Crises Under the National Banking System*. Government Printing Office, Washington.

Tallman, E. W. and Moen, J. R. (1998). Gold shocks, liquidity, and the United States economy during the National Banking Era. *Explorations in Economic History*, 35(4), 381–404.

Timberlake, R. H. (1993). *Monetary Policy in the United States*. University of Chicago Press, Chicago.

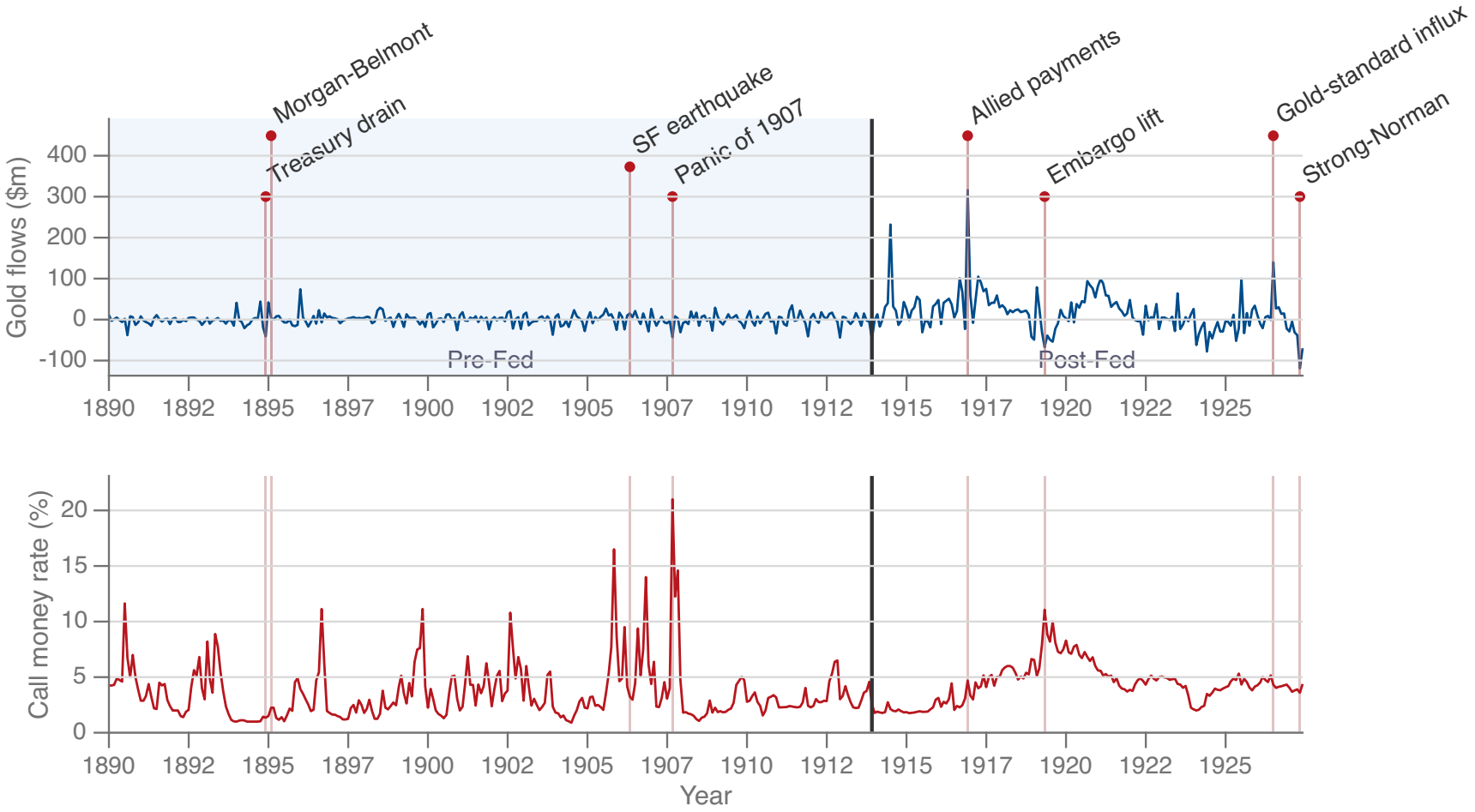
Uhlig, H. (2005). What are the effects of monetary policy on output. *Journal of Monetary Economics*, 52(2), 381–419.

Wicker, E. (2000). *Banking Panics of the Gilded Age*. Cambridge University Press, Cambridge.

A Data and Descriptive Statistics

Table 4 lists the six series and their sources, and Table 5 reports summary statistics over the estimation sample, 1890:01 to 1927:06. The five log-change series are shown in percent and the call money rate in percent. Two features are worth noting. The call money rate and the gold flow are both far from normal, with heavy right tails, high kurtosis, and, for the call rate, the twenty-one percent reading of the Panic of 1907. The gold flow has a standard deviation of about 33 million dollars, which is the unit we use to scale the impulse responses in the body. Figure 8 reproduces the series of Figure 1 on a full landscape page with each of the eight episodes labeled, so their timing within the sample can be read directly.

Figure 8: Gold flows and the money market, 1890 to 1927, with episodes labeled



Notes. The top panel plots the monthly change in the U.S. gold stock in millions of dollars, with the eight narrative episodes of Table 1 marked and labeled. The bottom panel plots the New York call money rate in percent, with the same episode dates marked. The heavy vertical line marks the Federal Reserve Act of December 1913 and shading marks the pre-Fed period.

Table 4: Variables and data sources

Variable	Description	Source
Industrial production	Monthly log change of the Miron-Romer index of industrial production, 1909 = 100	Miron and Romer (1990)
Call money rate	New York call money rate, percent, constructed from daily quotations	Fohlin (2025)
Equity return	Monthly log change of a stock price index	NBER Macrohistory, `M1125AUSM343NNBR`
USD/GBP	Monthly log change of the dollar-sterling rate	`USUKFXUKM`
Price	Monthly log change of the general price index	NBER Macrohistory, `M04051USM324NNBR`
Gold flows	Monthly change in the stock of gold held by the Treasury and the Federal Reserve, end of month, millions of dollars	NBER Macrohistory, `M1437AUSM144NNBR` and `M1437BUSM144NNBR`

Notes. The gold series is the change in the monetary gold stock rather than a series on gold imports and exports, so it reflects domestic additions to monetary gold as well as international movements. The two gold series are spliced at the end of 1914.

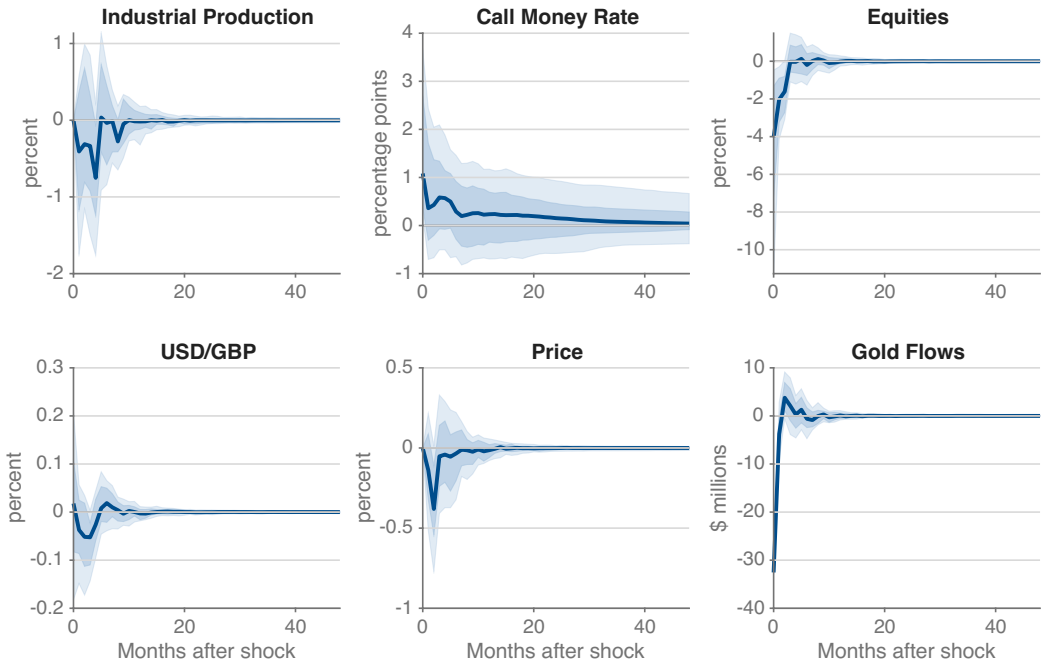
Table 5: Descriptive statistics, 1890:01 to 1927:06

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Industrial production (% chg)	0.14	3.40	−9.39	15.32	0.00	3.98
Call money rate (%)	3.86	2.35	0.90	21.00	2.23	12.41
Equity return (% chg)	0.26	3.40	−11.58	9.60	−0.34	3.32
USD/GBP (% chg)	0.00	1.18	−8.50	9.85	0.01	29.69
Price (% chg)	0.18	0.91	−3.82	3.23	−0.19	4.46
Gold flows ($ millions)	6.10	32.55	−119.00	316.00	3.03	26.67

B Impulse Responses by Episode

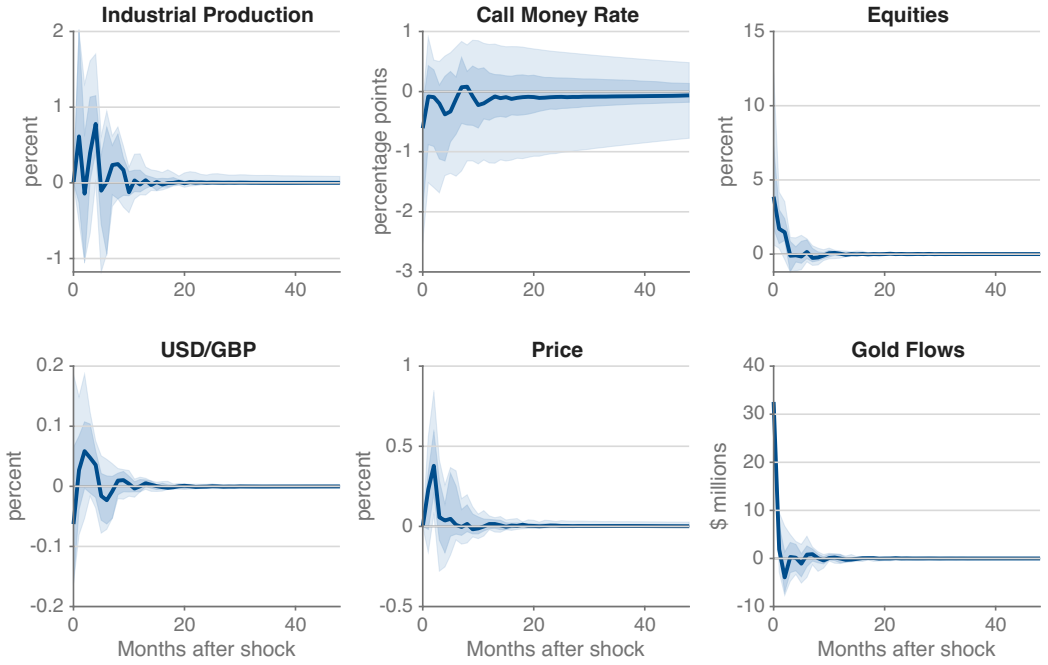
Figures 9 through 16 report the impulse responses conditioned on each of the eight episodes, discussed in Section 5.2. Each is estimated on a local window around the event, so the responses differ from one episode to the next, with the pre-Fed episodes showing large and variable liquidity effects and the post-Fed episodes muted responses near zero. Each response is scaled to a gold inflow of one standard deviation, and outflow episodes are shown as outflows. Solid lines are posterior medians and shaded regions are the 68 and 90 percent pointwise credible bands. The pre-Fed episodes, and the two whose windows are shortest, carry visibly wider bands.

Figure 9: Impulse responses, Treasury gold-reserve crisis (December 1894), gold outflow



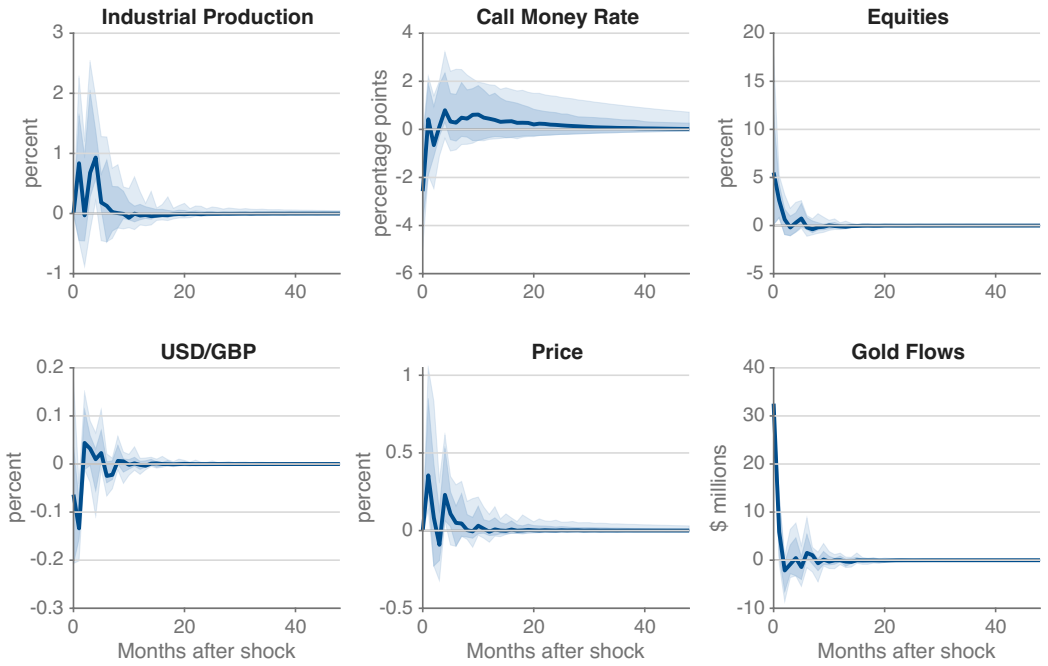
Notes. Response to a gold outflow of one standard deviation, with 68 and 90 percent bands.

Figure 10: Impulse responses, Morgan-Belmont syndicate (February 1895), gold inflow



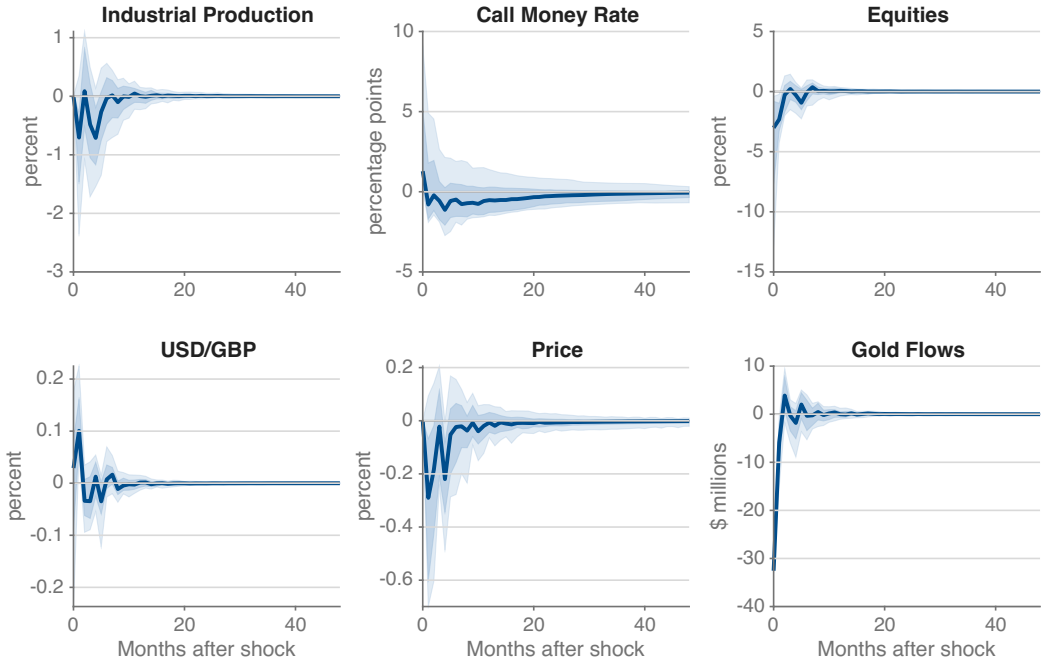
Notes. Response to a gold inflow of one standard deviation, with 68 and 90 percent bands.

Figure 11: Impulse responses, San Francisco earthquake (May 1906), gold inflow



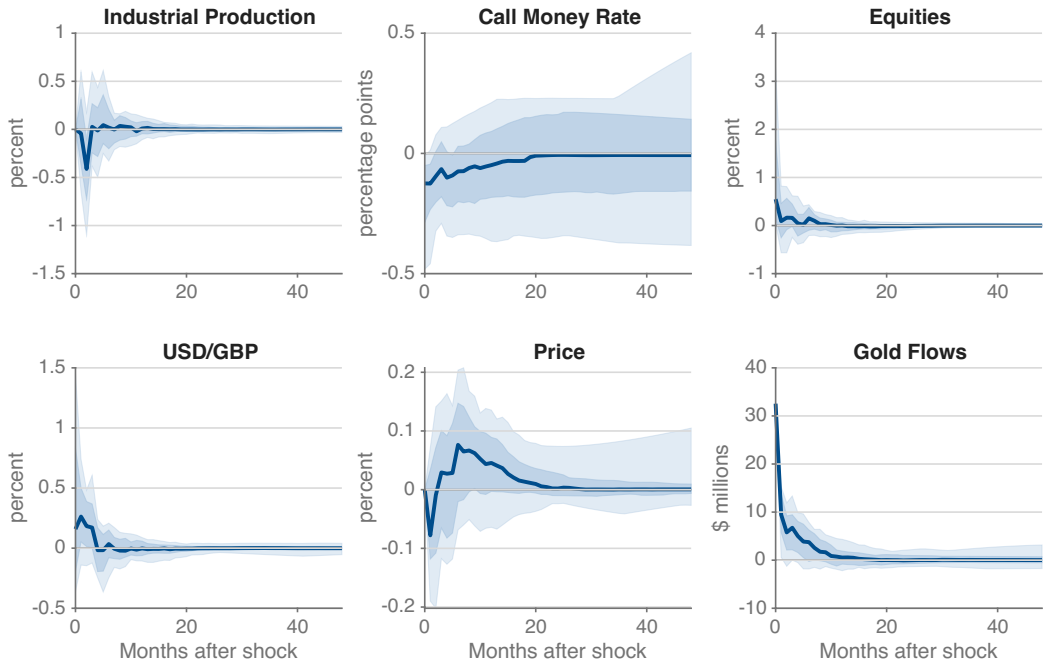
Notes. Response to a gold inflow of one standard deviation, with 68 and 90 percent bands.

Figure 12: Impulse responses, Panic of 1907 drain (September 1907), gold outflow



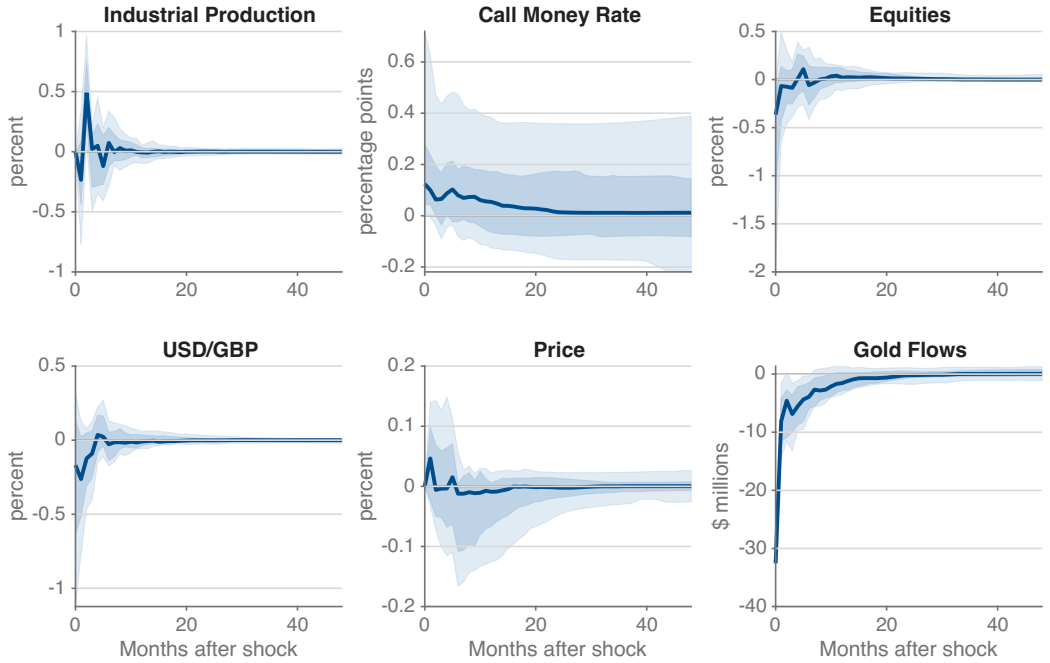
Notes. Response to a gold outflow of one standard deviation, with 68 and 90 percent bands.

Figure 13: Impulse responses, Allied wartime payments (December 1916), gold inflow



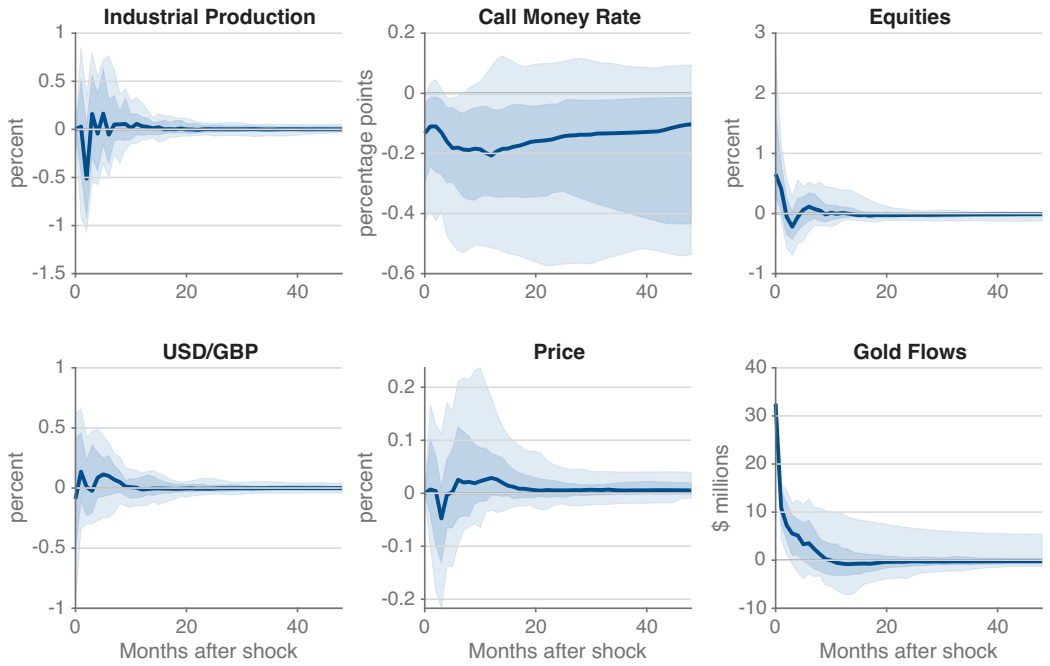
Notes. Response to a gold inflow of one standard deviation, with 68 and 90 percent bands.

Figure 14: Impulse responses, gold-export embargo lift (May 1919), gold outflow



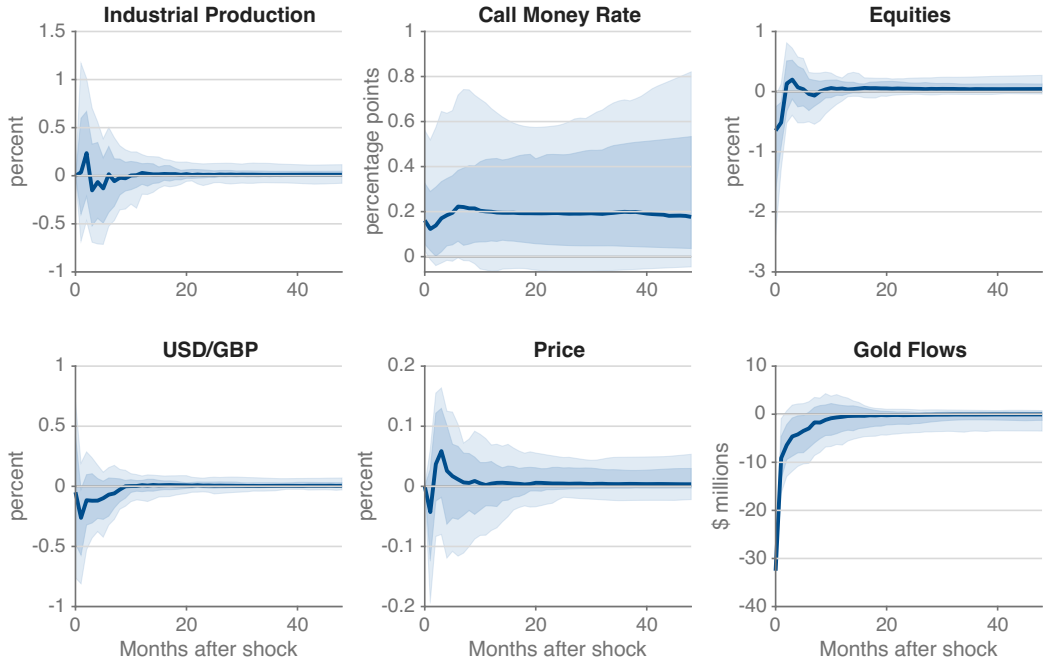
Notes. Response to a gold outflow of one standard deviation, with 68 and 90 percent bands.

Figure 15: Impulse responses, restored gold-standard influx (July 1926), gold inflow



Notes. Response to a gold inflow of one standard deviation, with 68 and 90 percent bands.

Figure 16: Impulse responses, Strong-Norman episode (May 1927), gold outflow

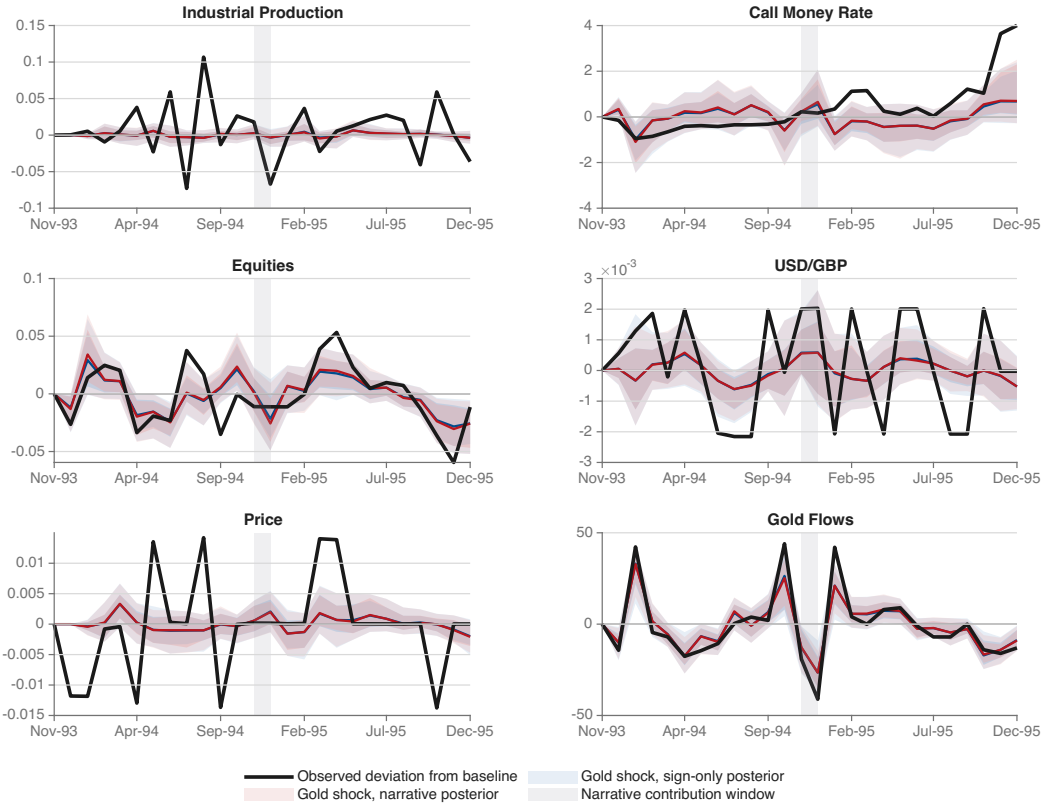


Notes. Response to a gold outflow of one standard deviation, with 68 and 90 percent bands.

C Additional Historical Decompositions

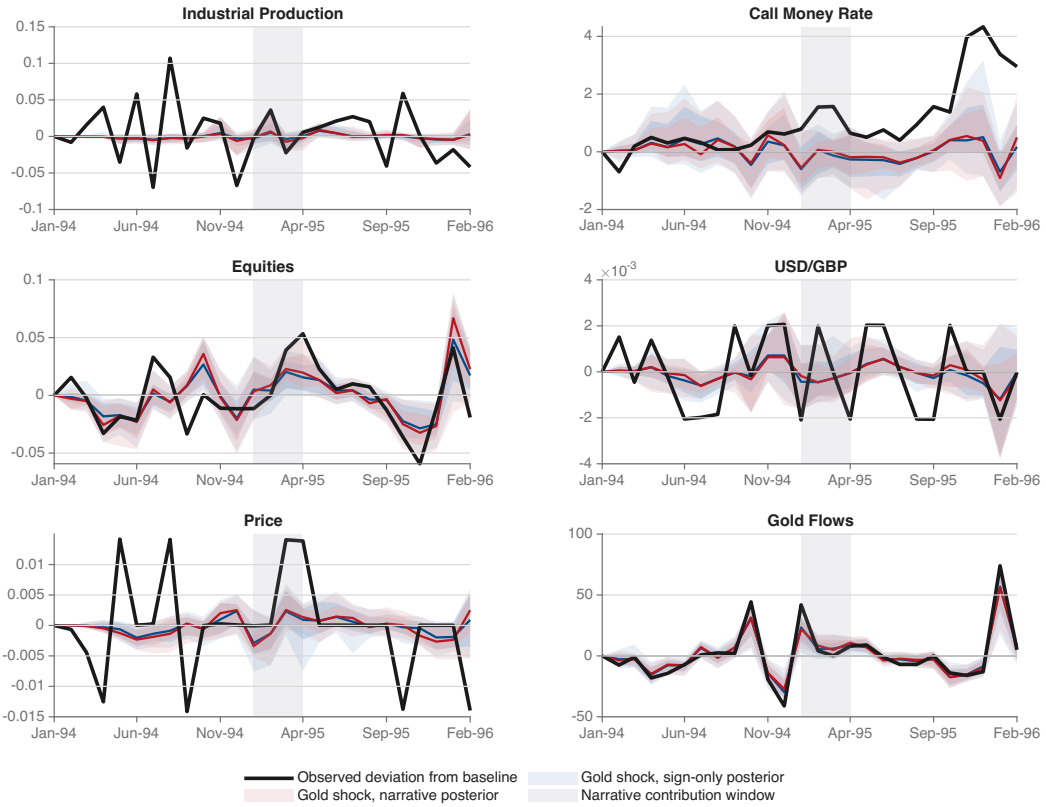
The body reports historical decompositions for the Panic of 1907 and the Allied wartime payments. Figures 17 through 22 report the remaining six, one for each of the other episodes. The reading is the same as in the body. The black line is the observed deviation from the deterministic baseline, the blue band is the gold-shock contribution under the sign-only posterior, and the red band is the contribution under the narrative posterior, each with 68 and 90 percent bands, and shading marks the contribution window.

Figure 17: Historical decomposition, Treasury gold-reserve crisis (December 1894)



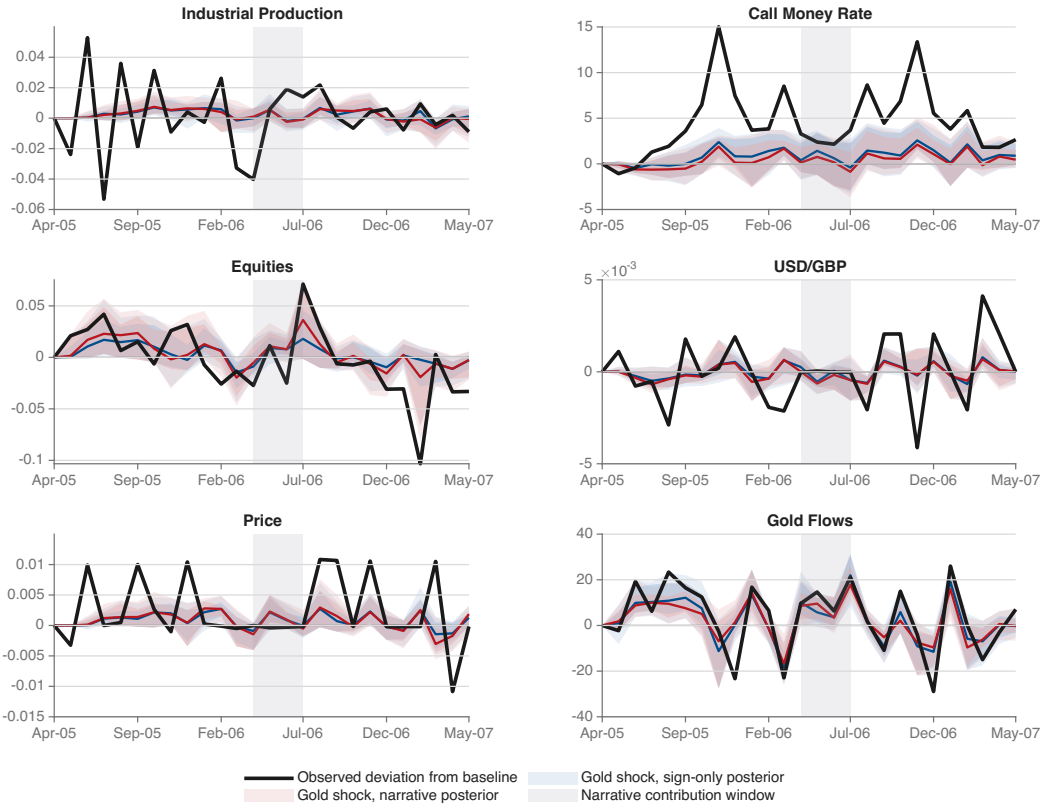
Notes. See the notes to Figure 4.

Figure 18: Historical decomposition, Morgan-Belmont syndicate (February 1895)



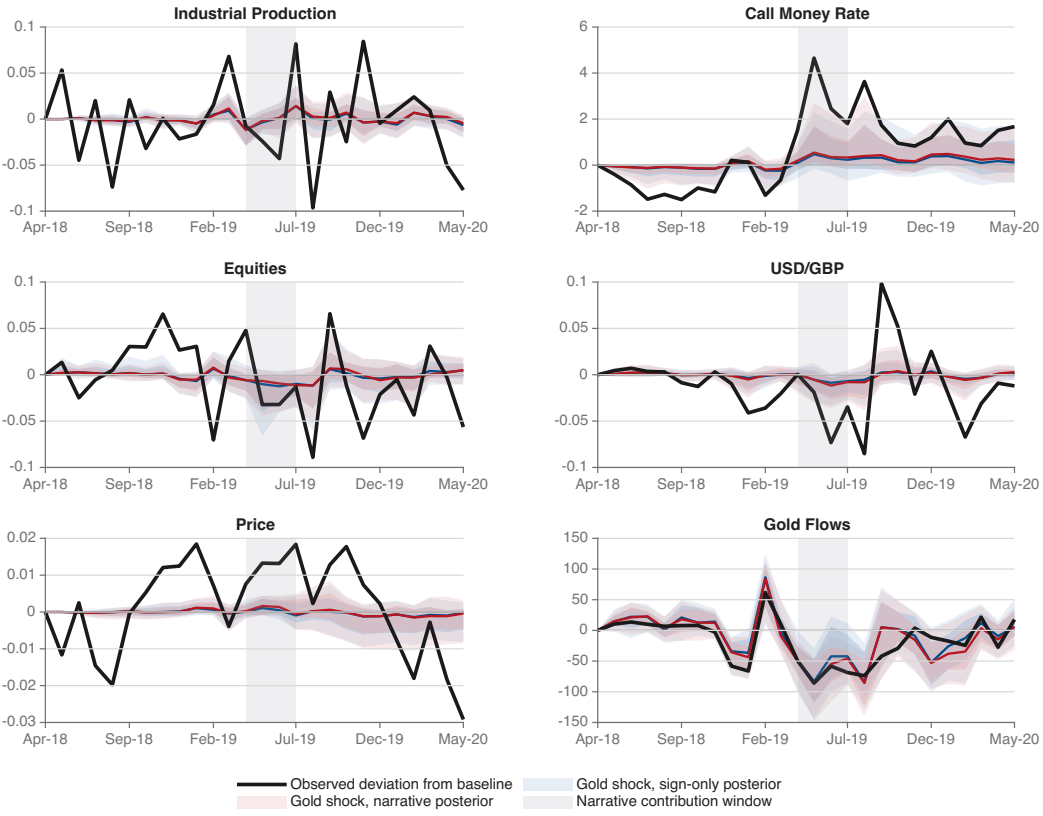
Notes. See the notes to Figure 4.

Figure 19: Historical decomposition, San Francisco earthquake (May 1906)



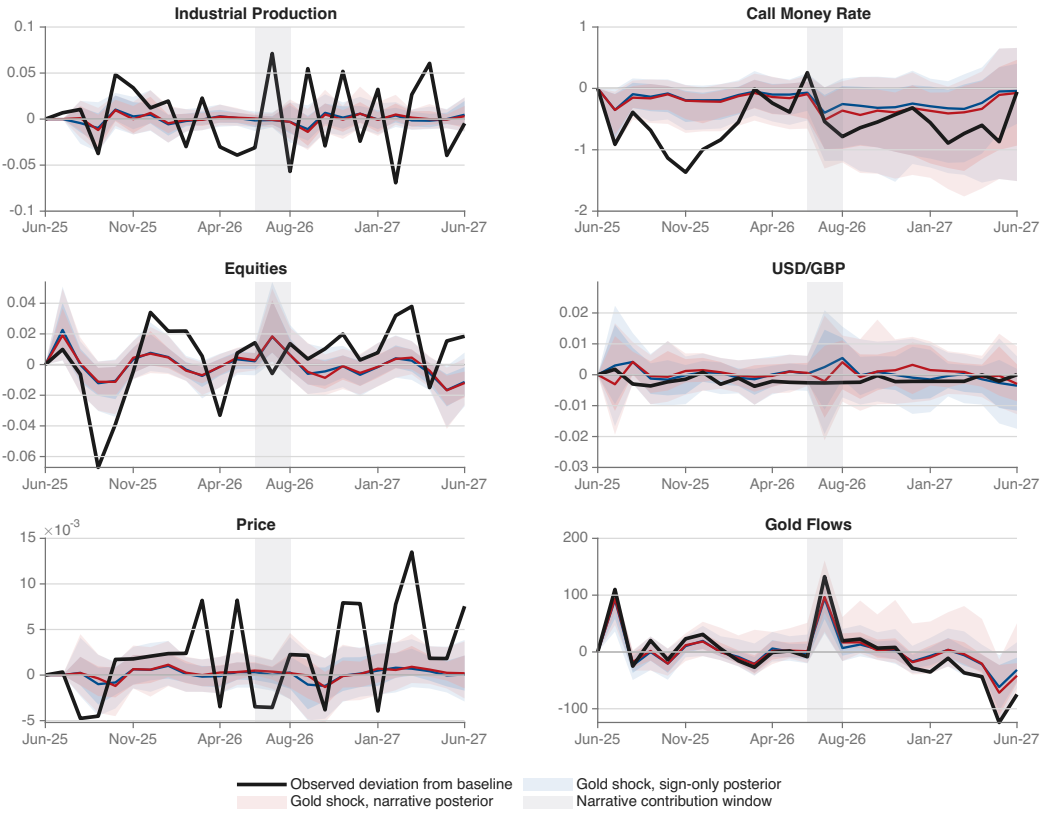
Notes. See the notes to Figure 4.

Figure 20: Historical decomposition, gold-export embargo lift (May 1919)



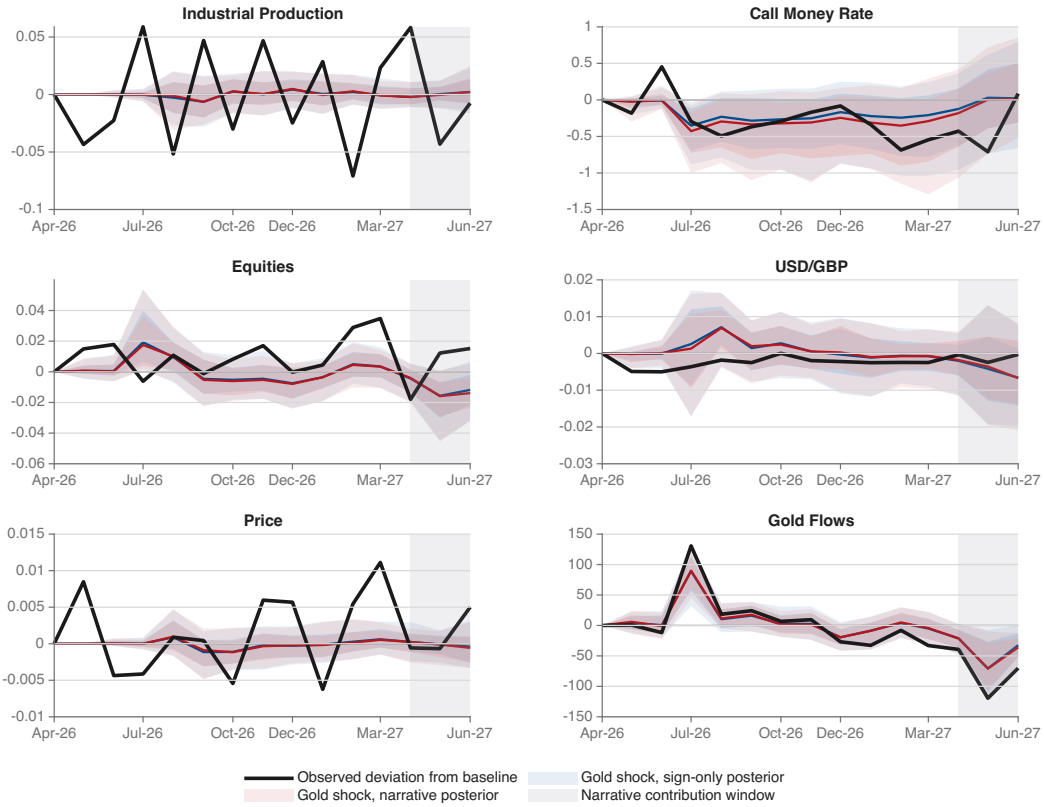
Notes. See the notes to Figure 4.

Figure 21: Historical decomposition, restored gold-standard influx (July 1926)



Notes. See the notes to Figure 4.

Figure 22: Historical decomposition, Strong-Norman episode (May 1927)



Notes. See the notes to Figure 4.